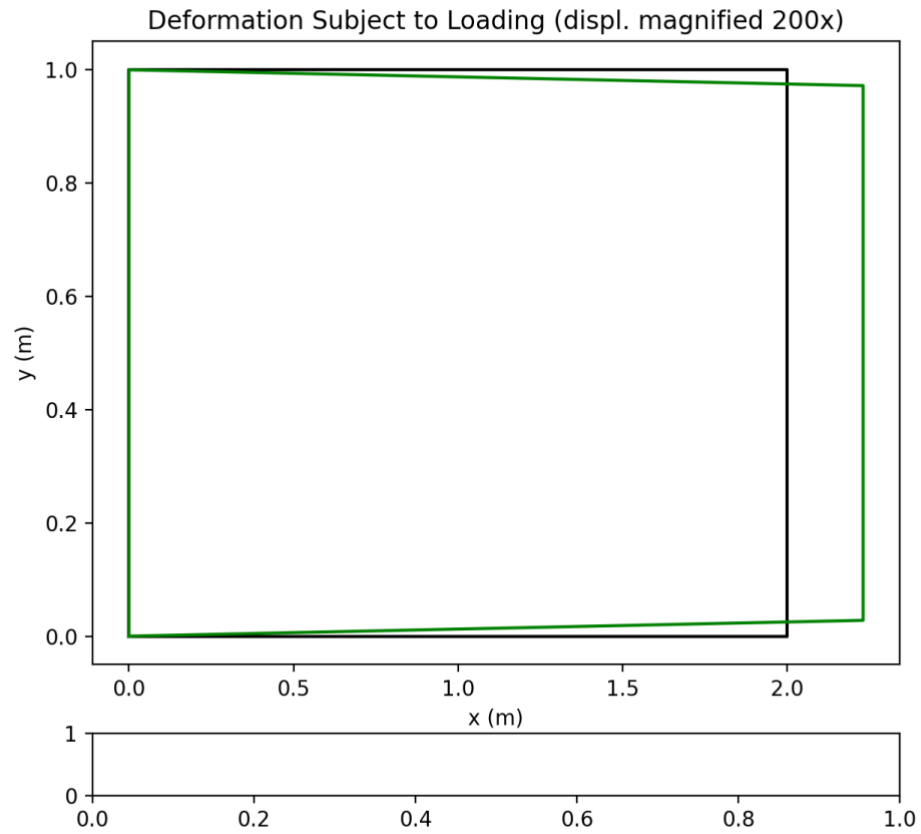
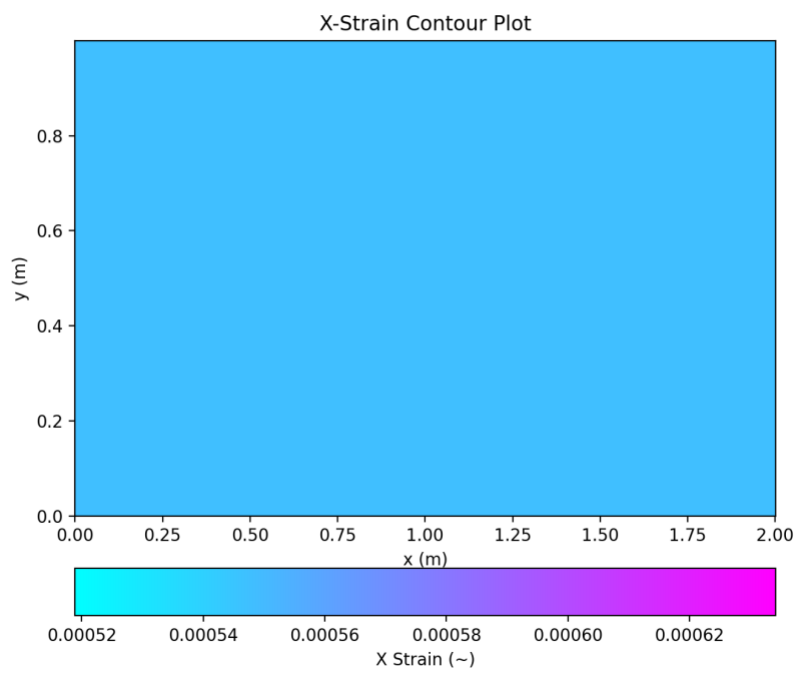
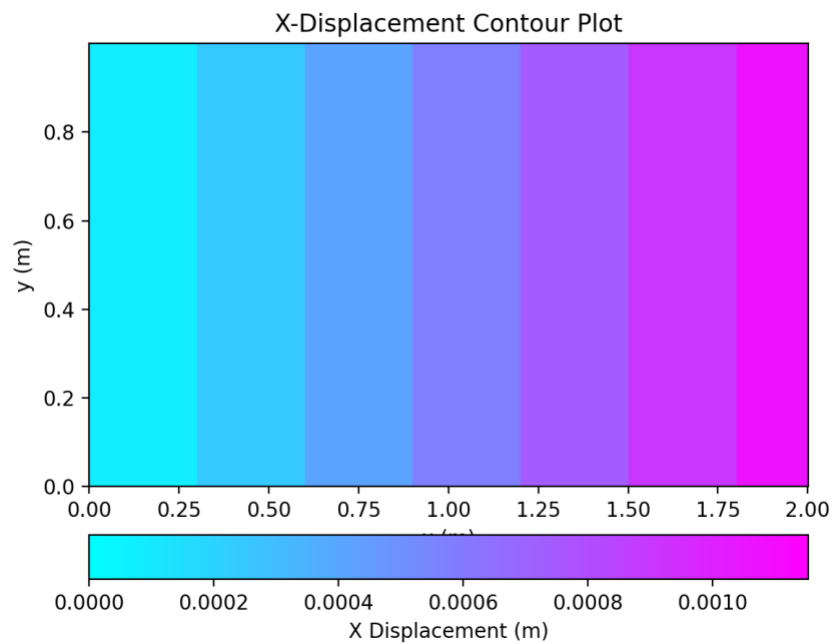
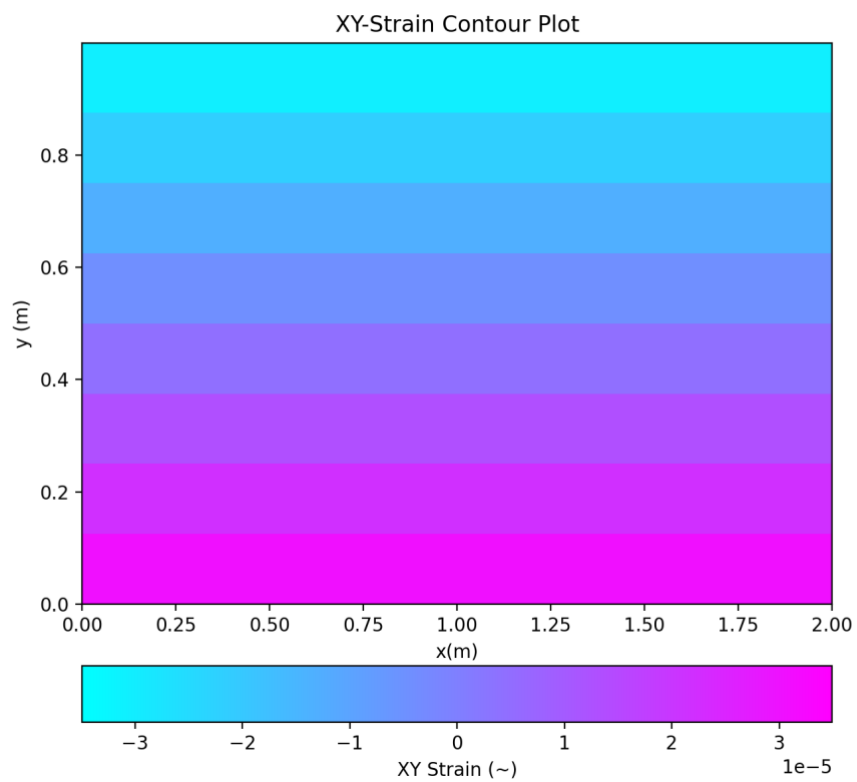
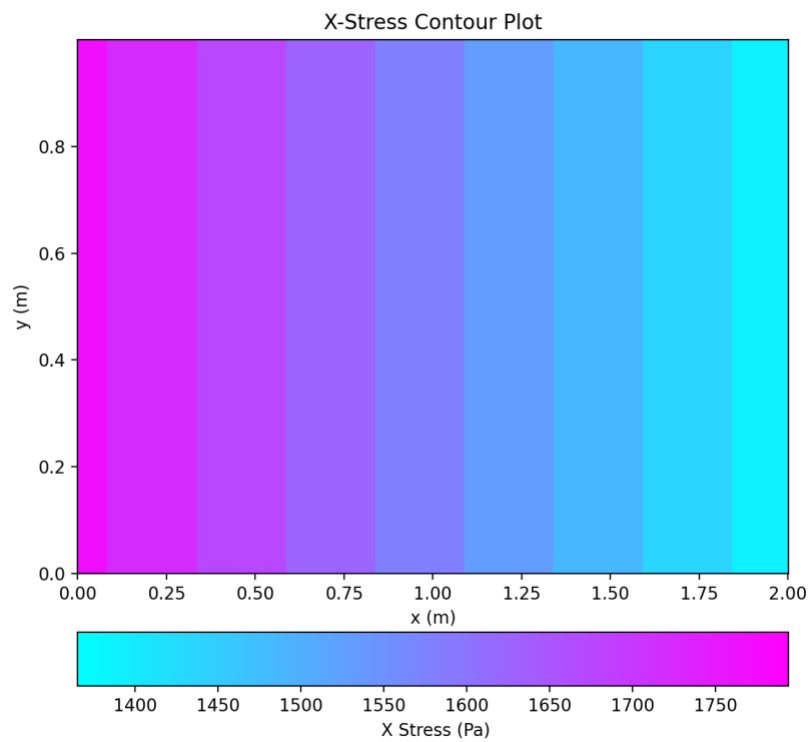


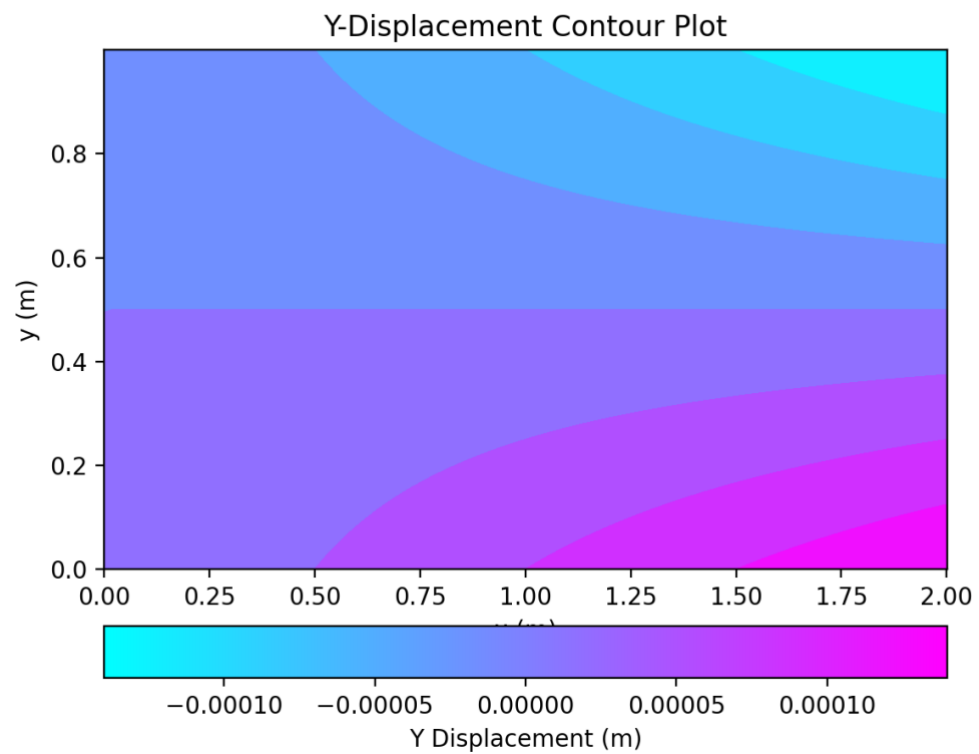
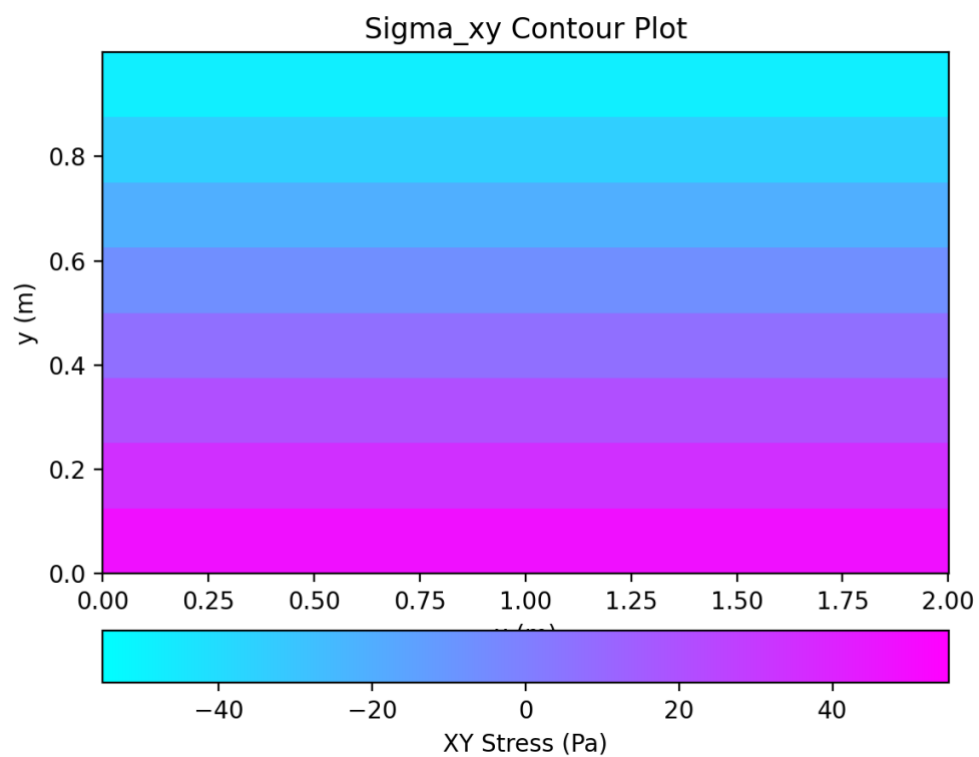
FEA ANALYSIS (MESH STUDY)

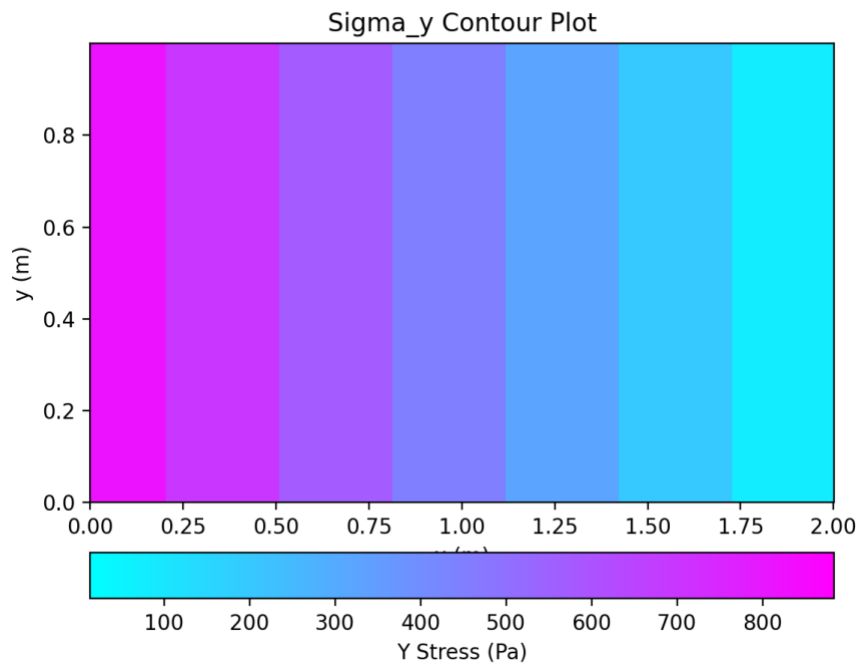
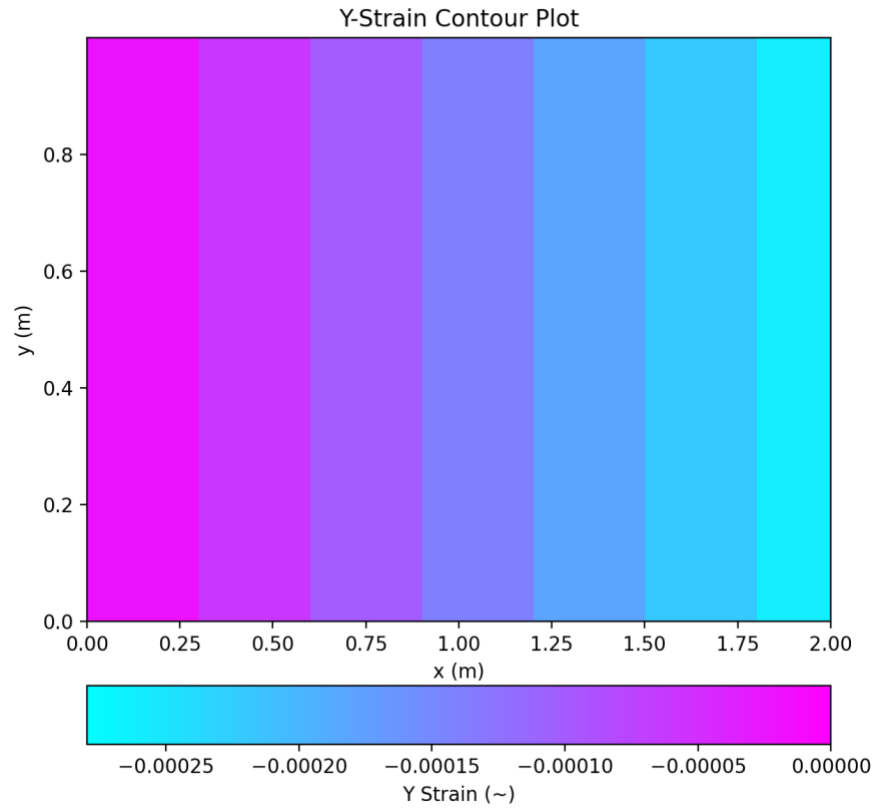
Mesh #1: 1x1

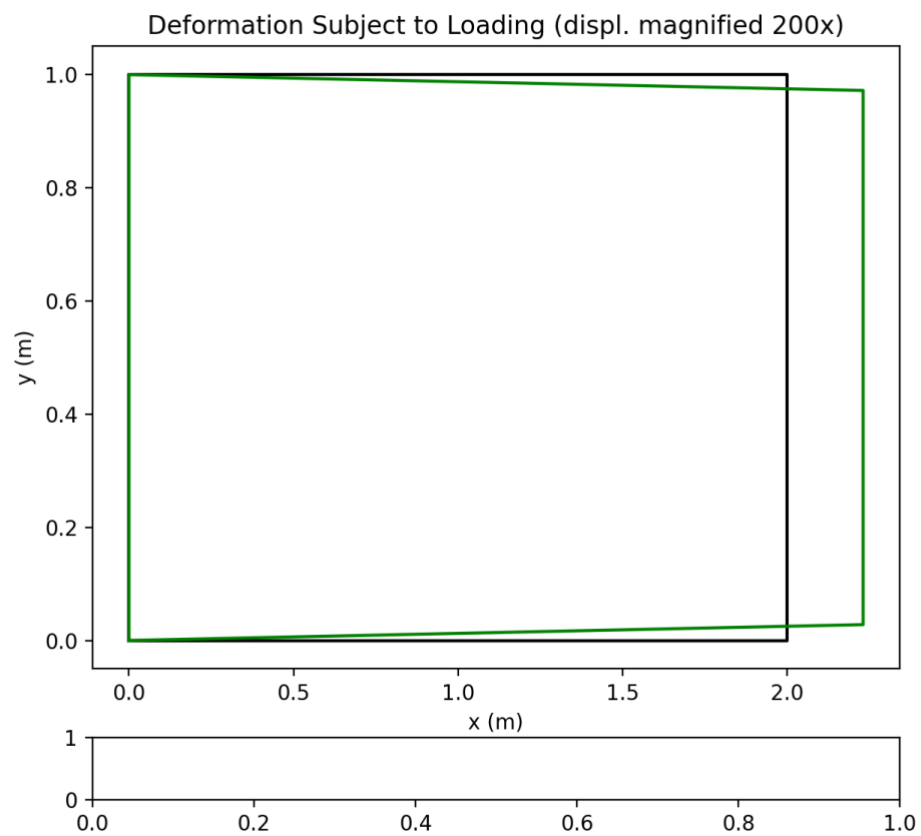




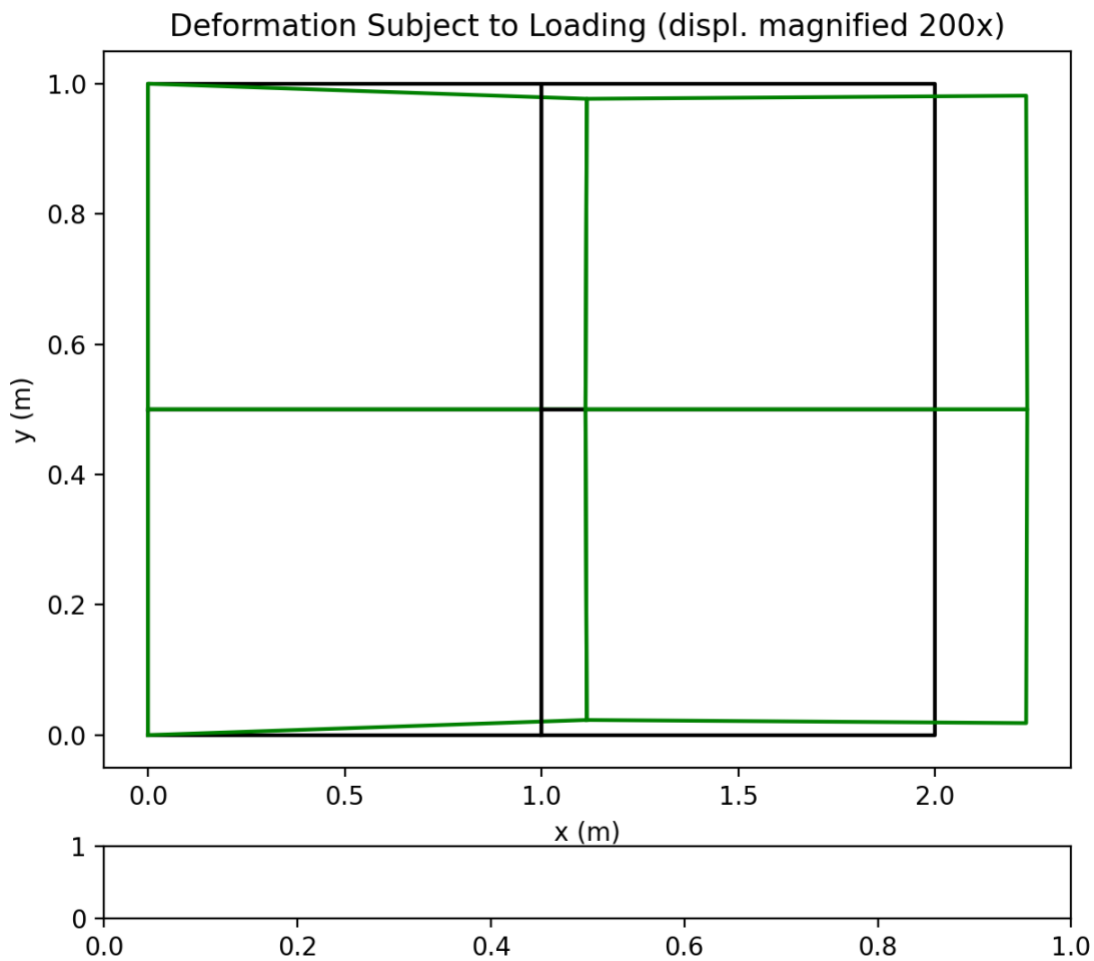


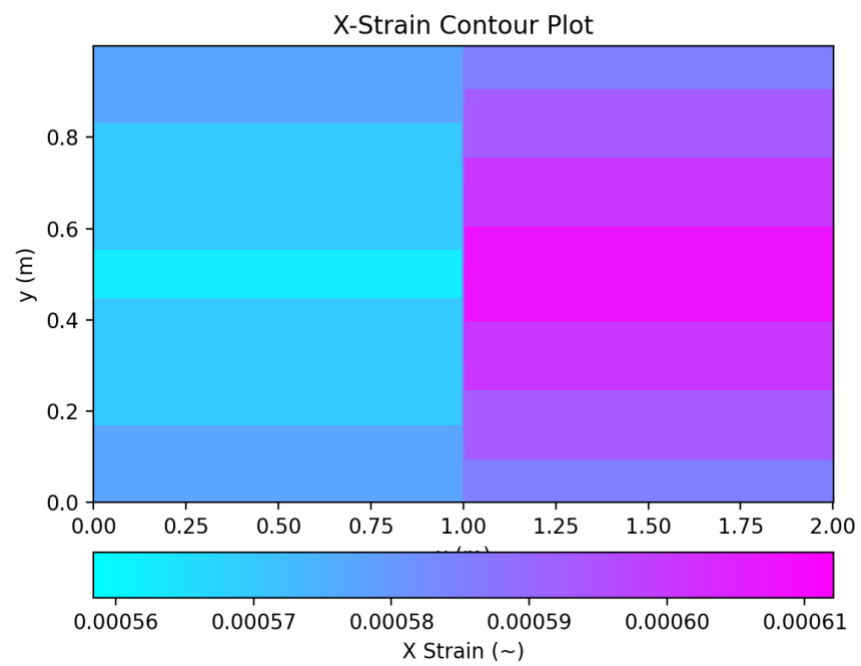
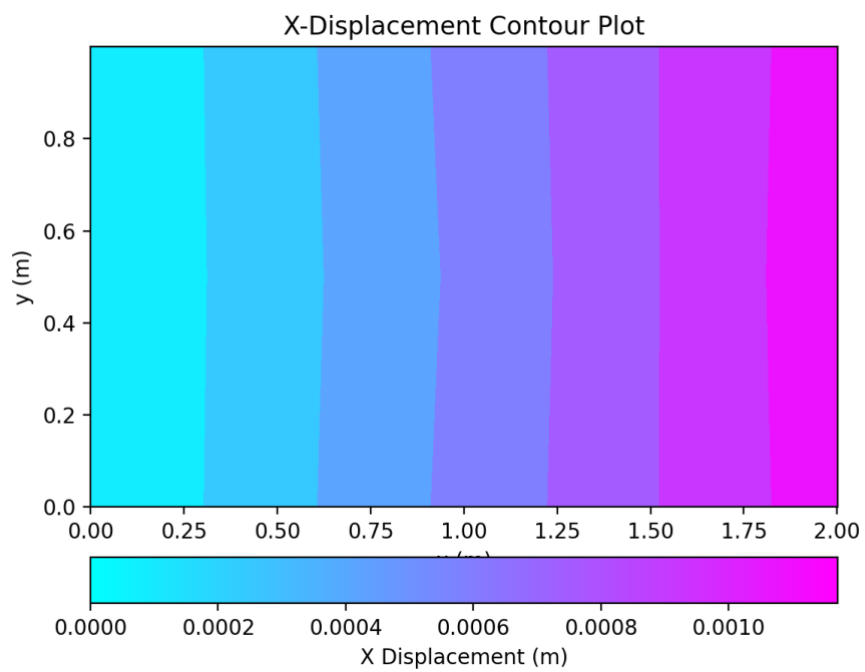


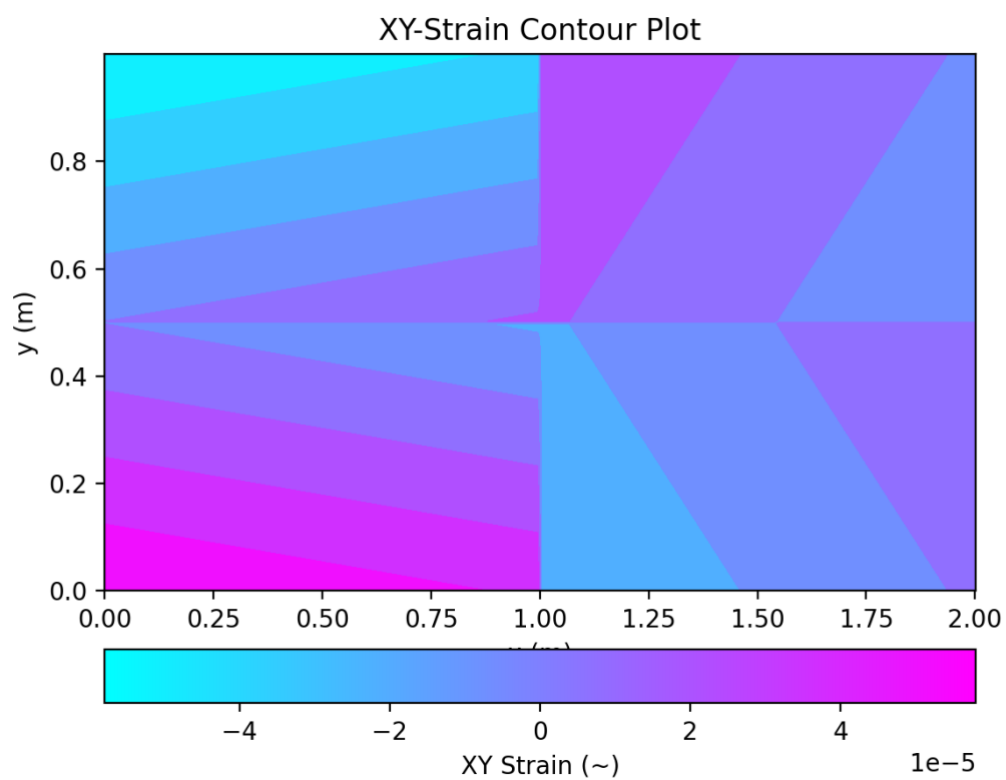
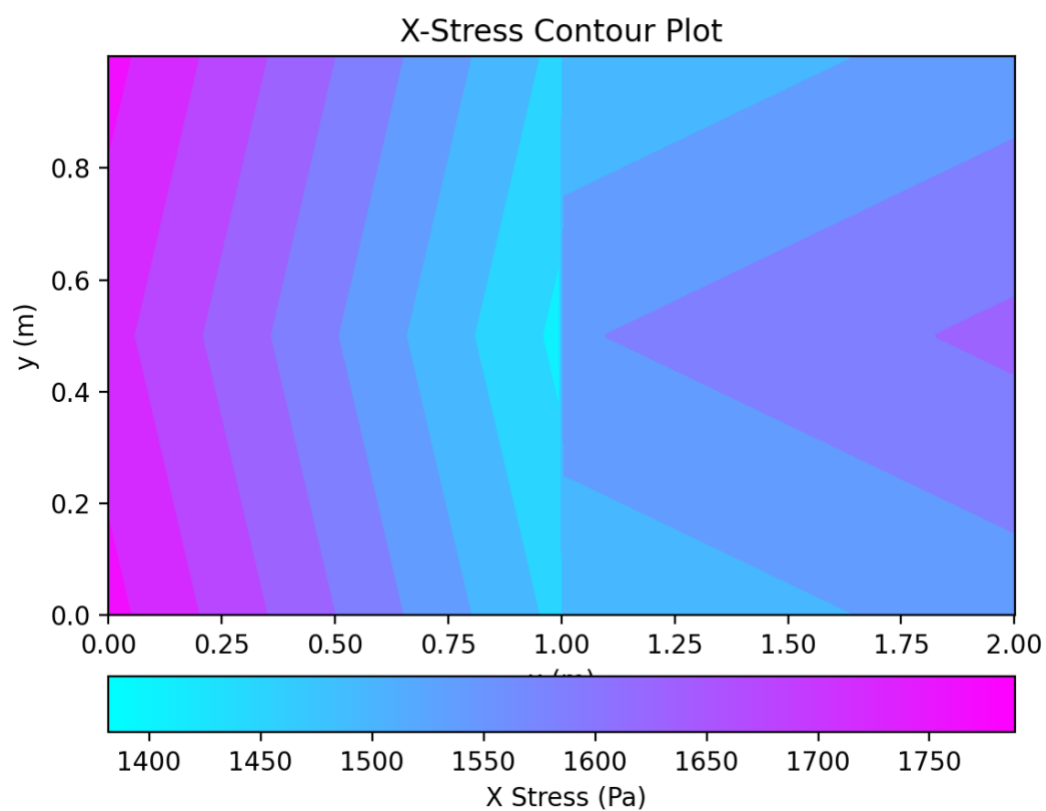


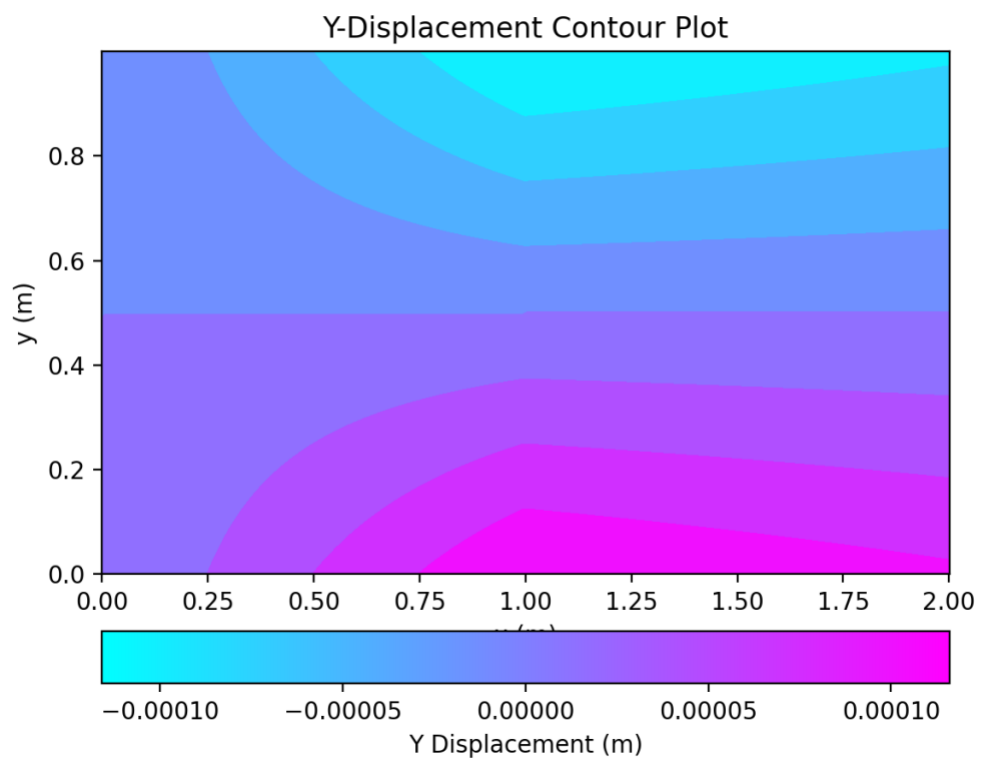
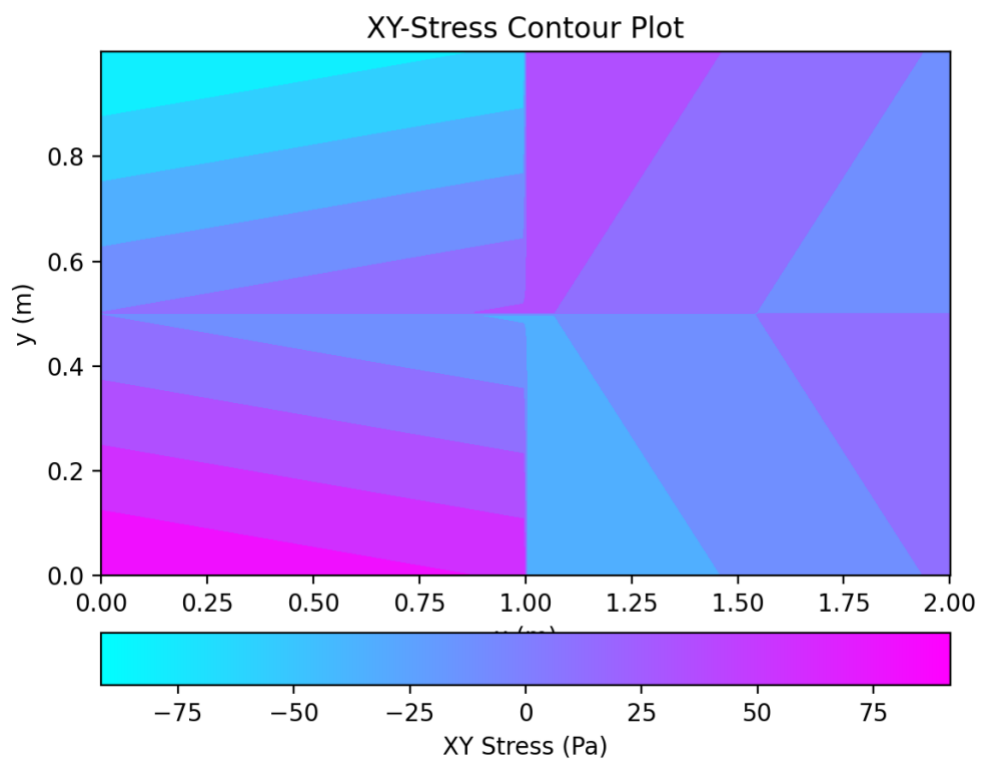


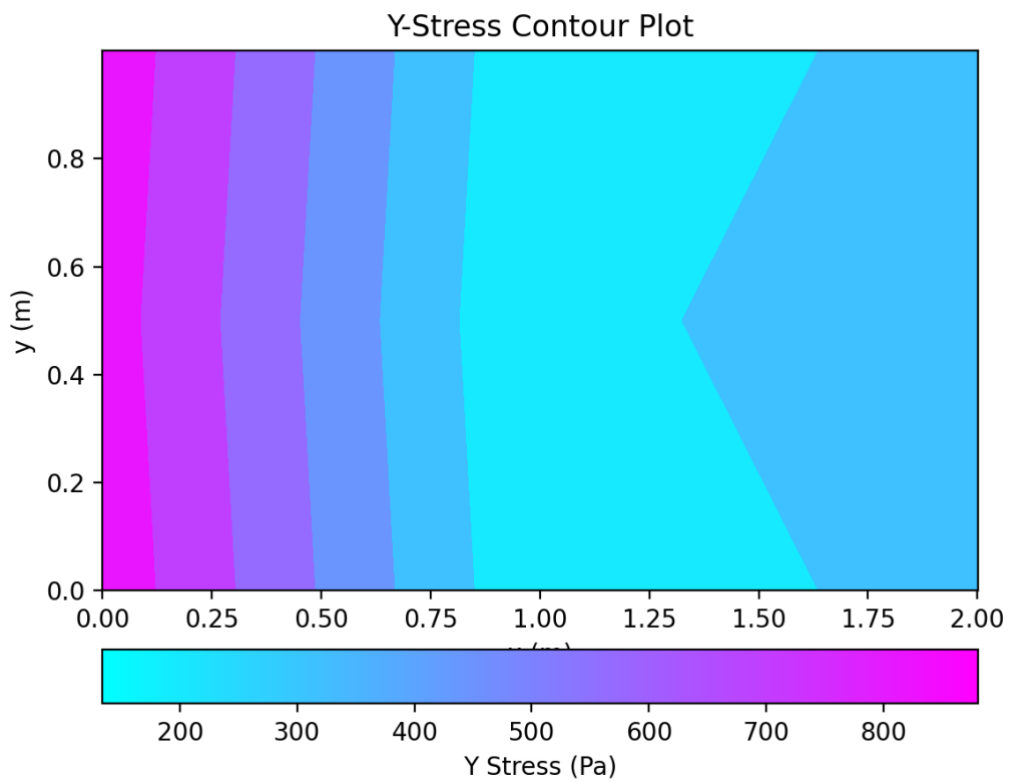
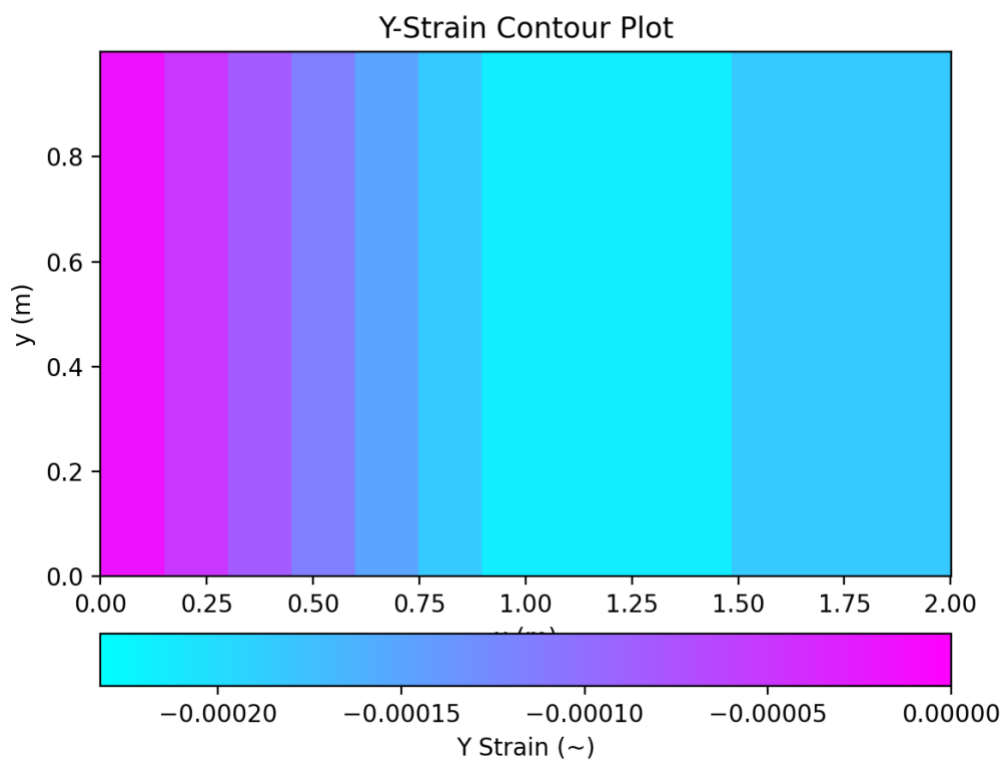
Mesh #2: 2x2



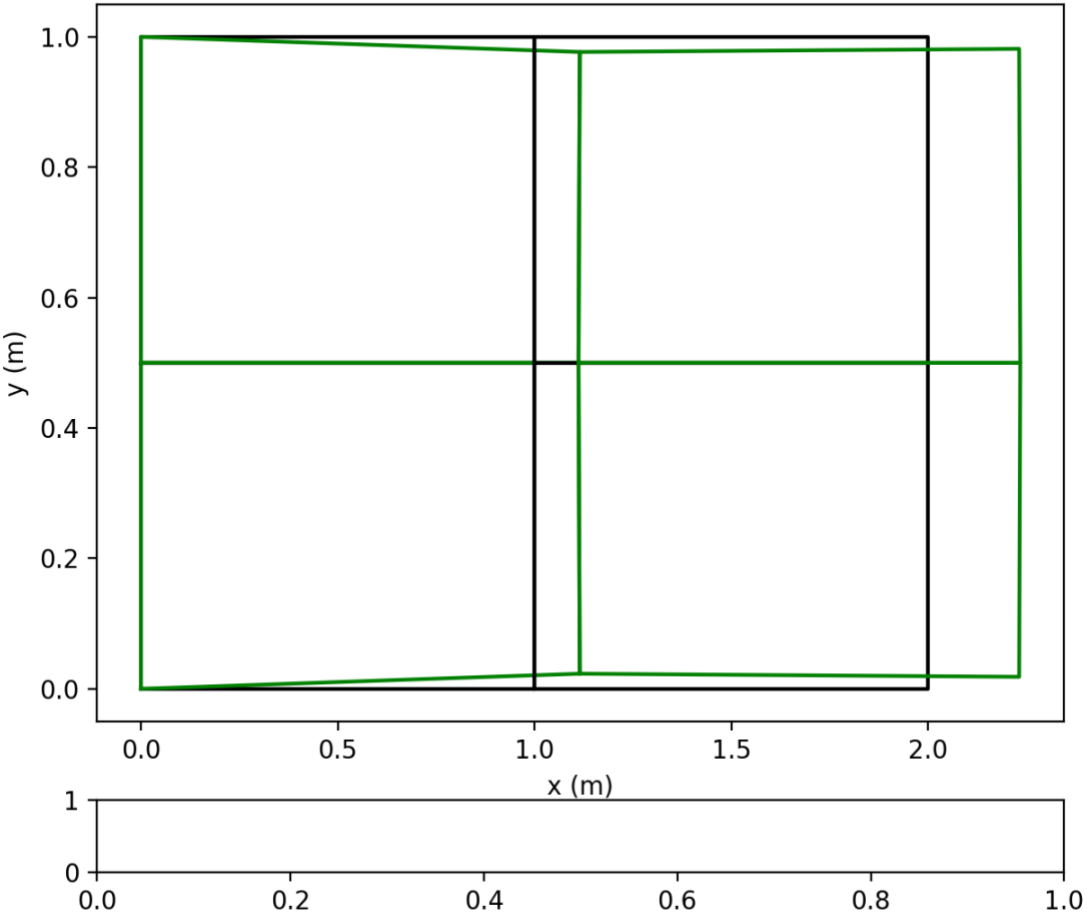




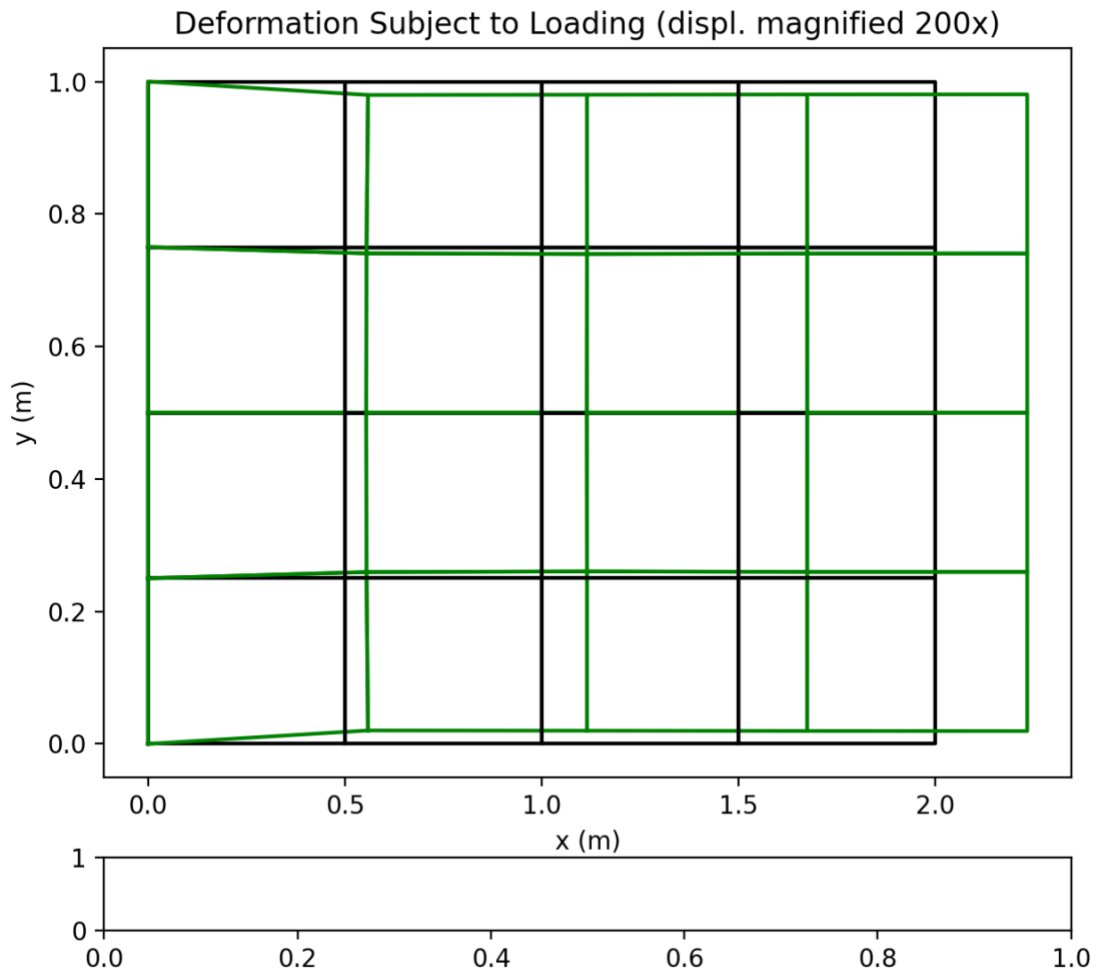


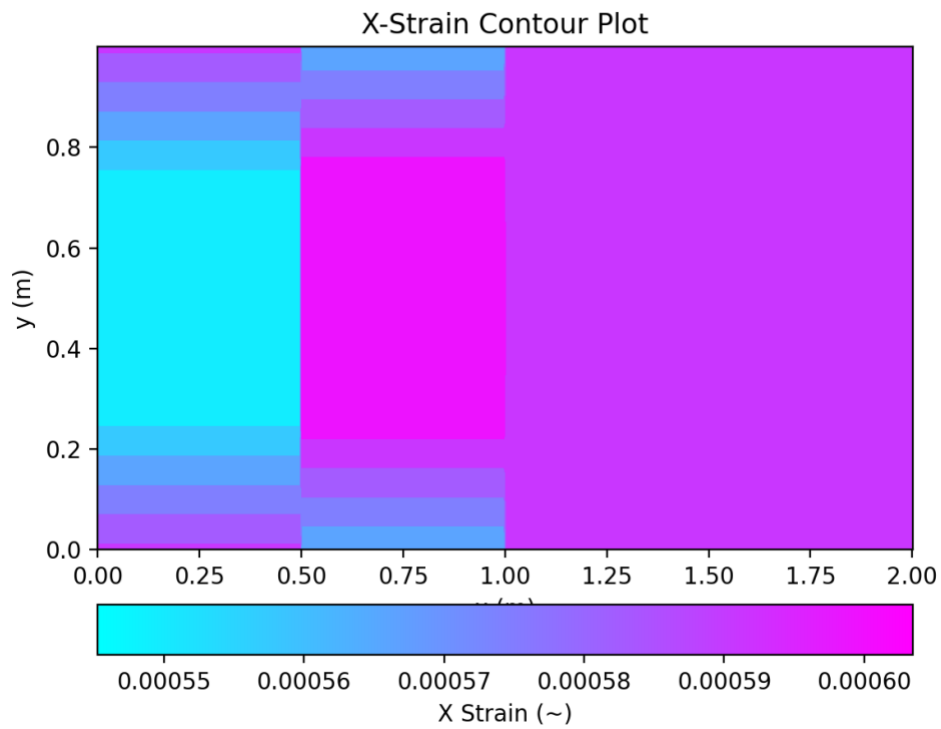
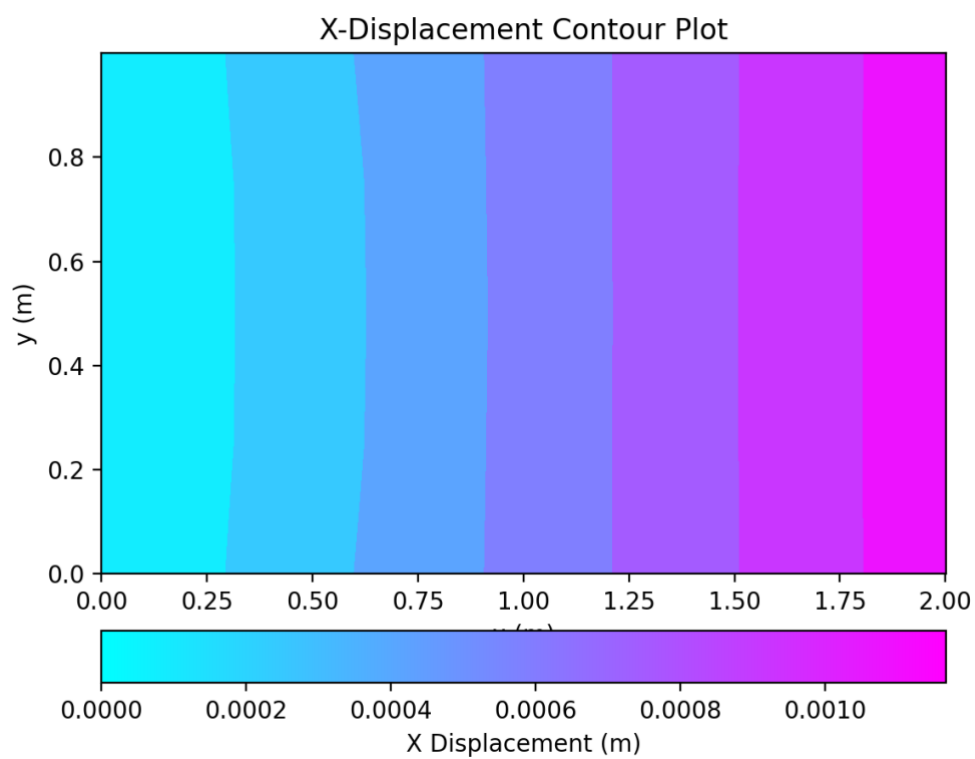


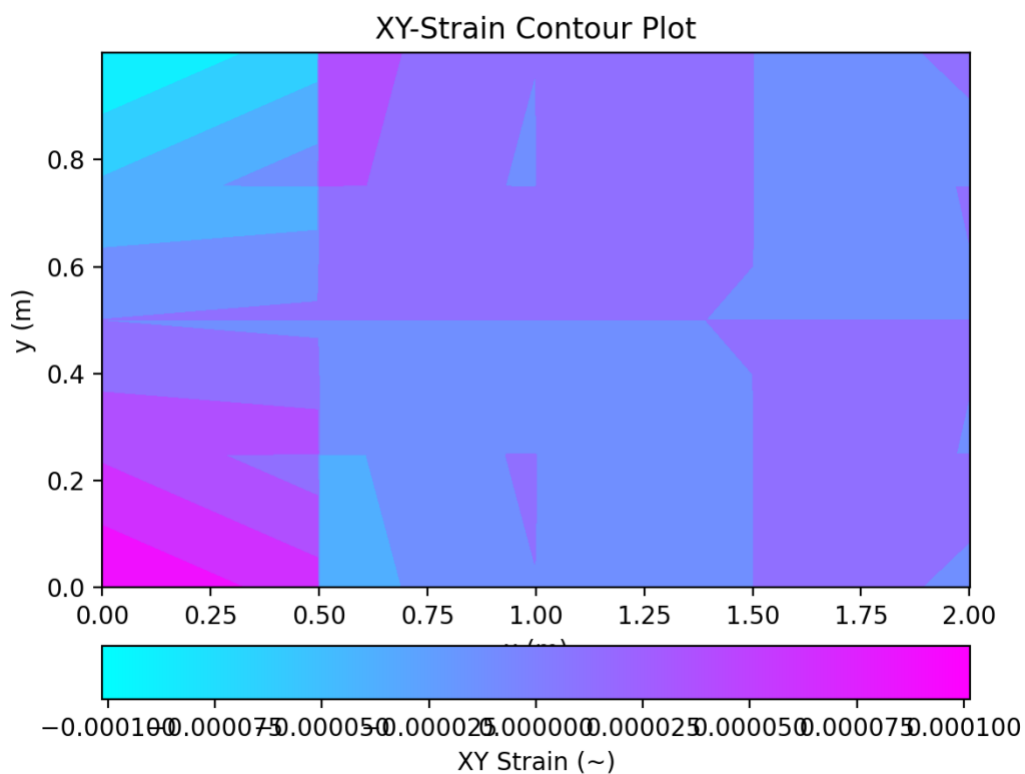
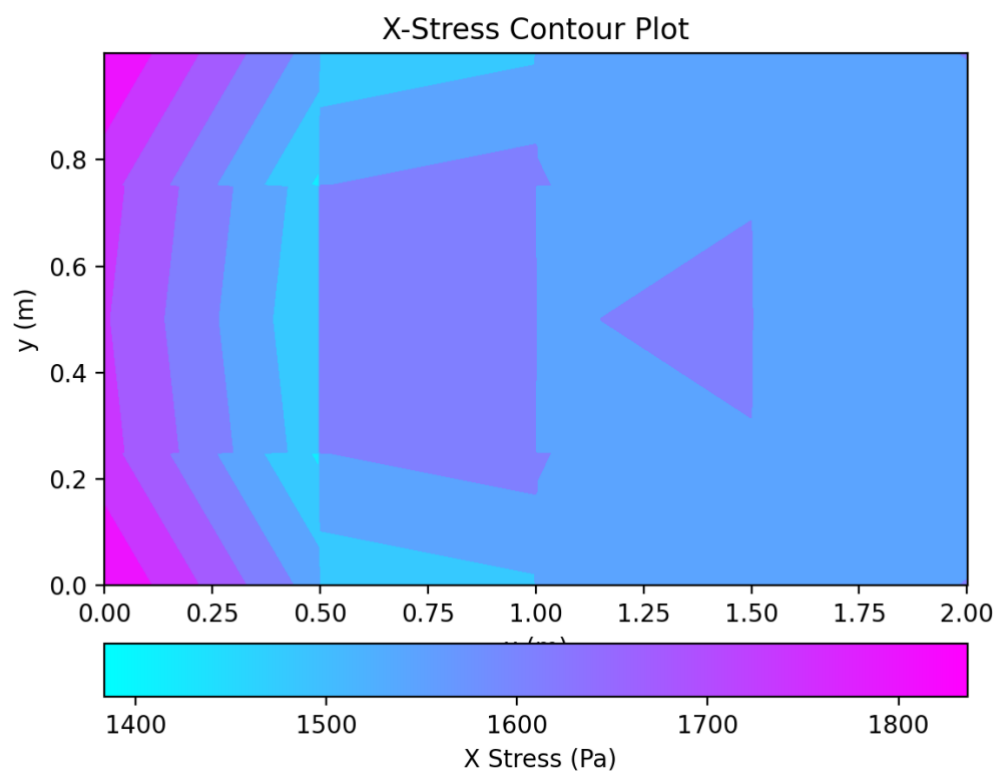
Deformation Subject to Loading (displ. magnified 200x)

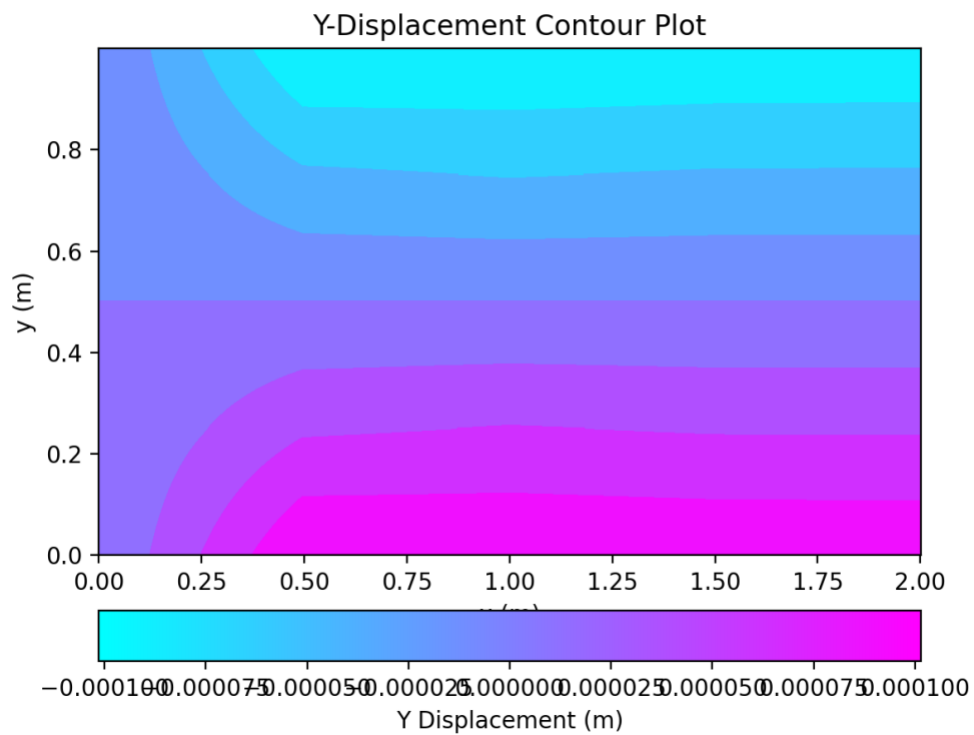
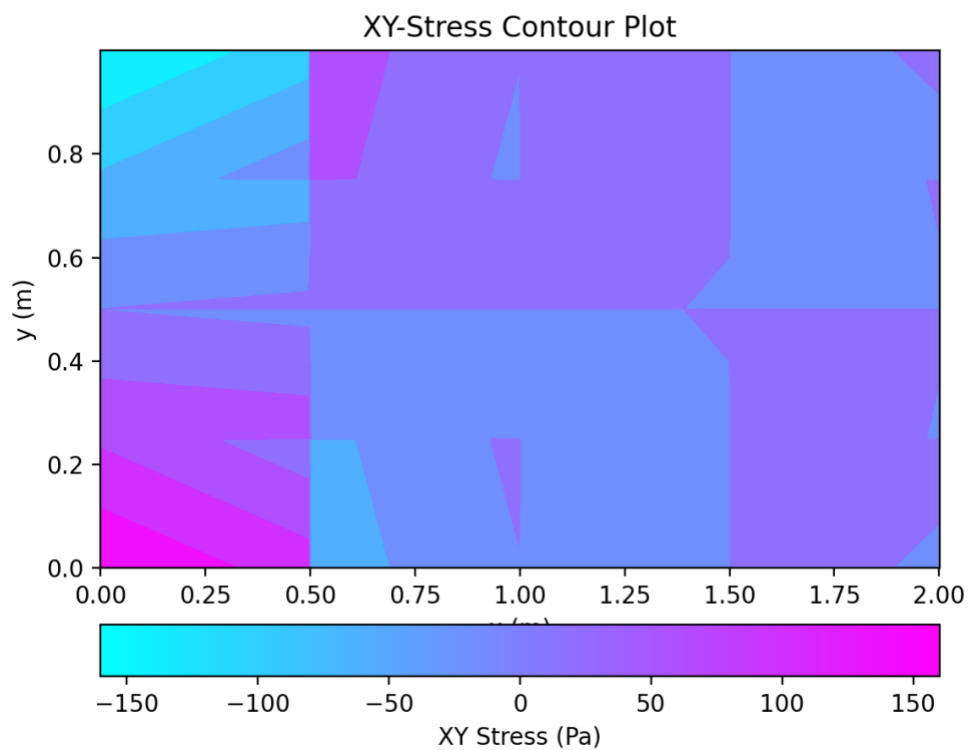


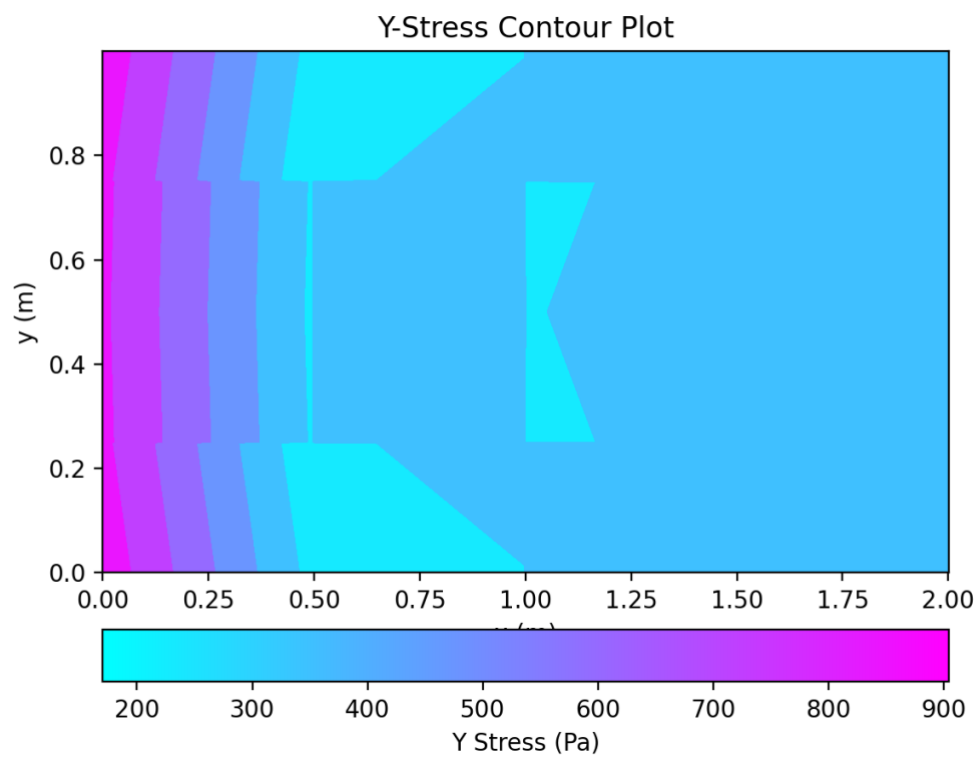
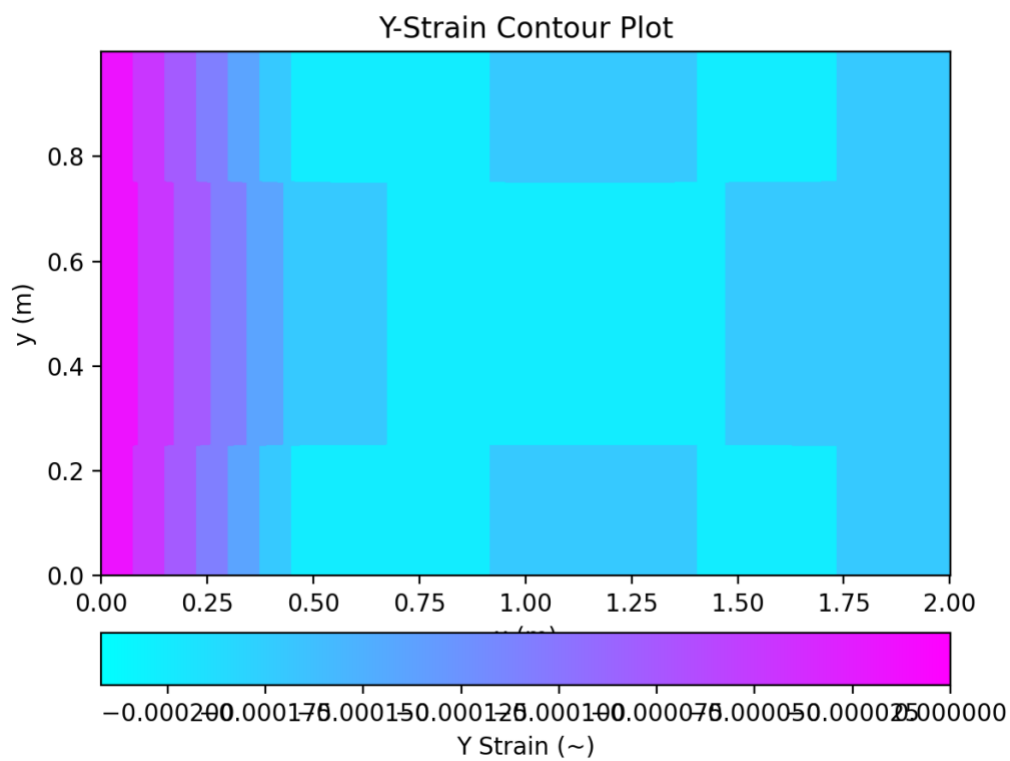
Mesh 3: 4x4



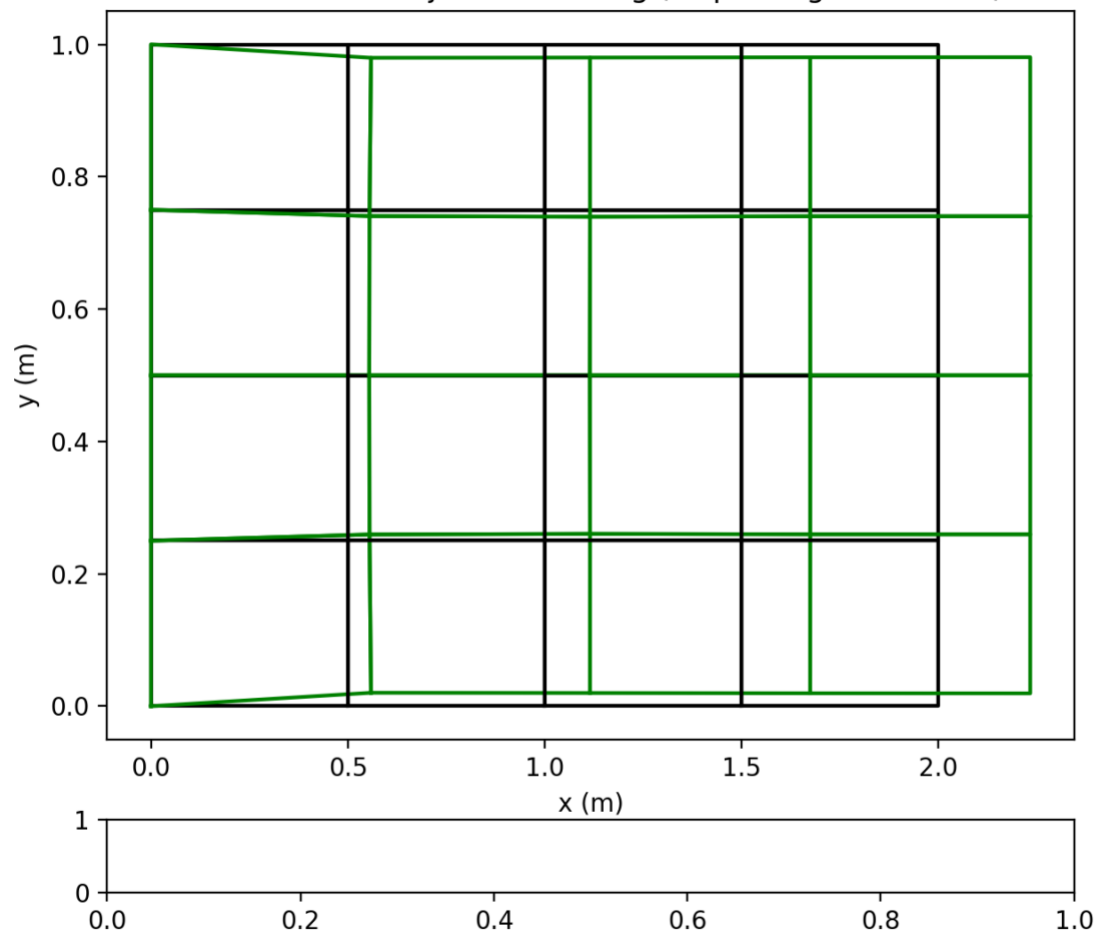




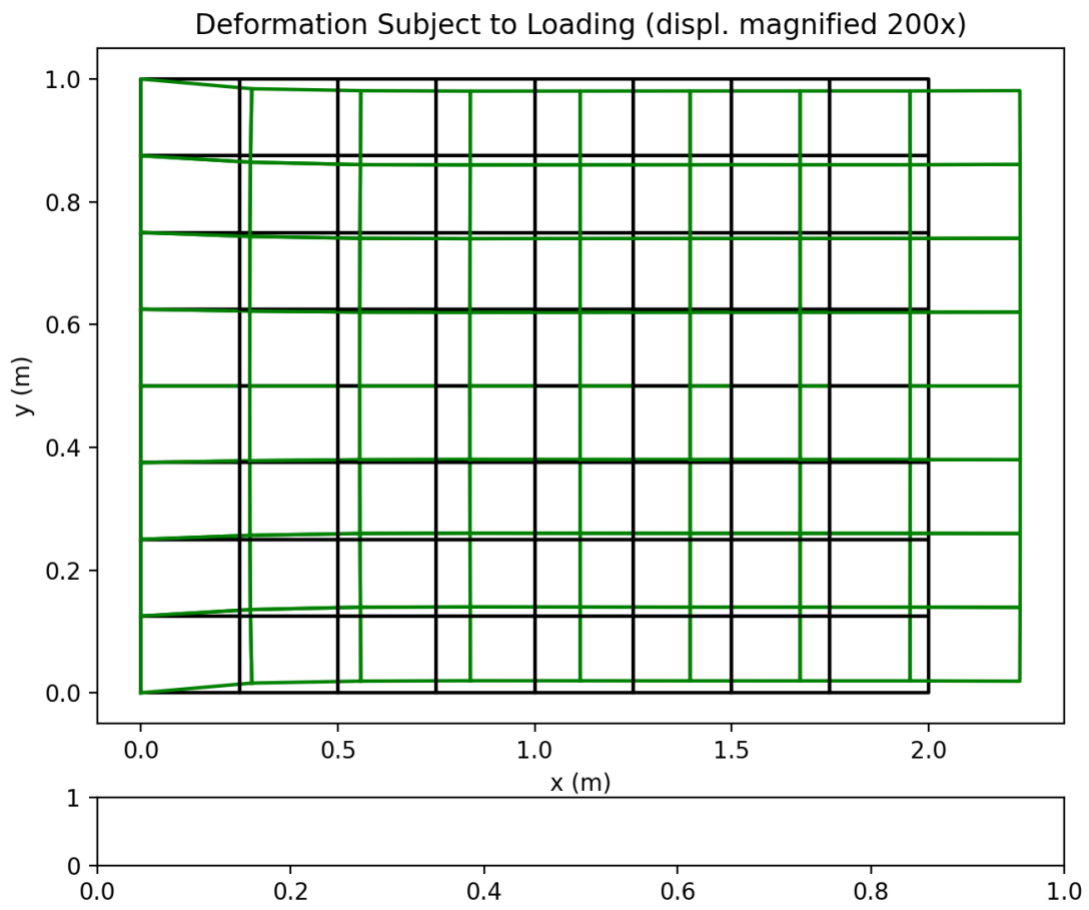


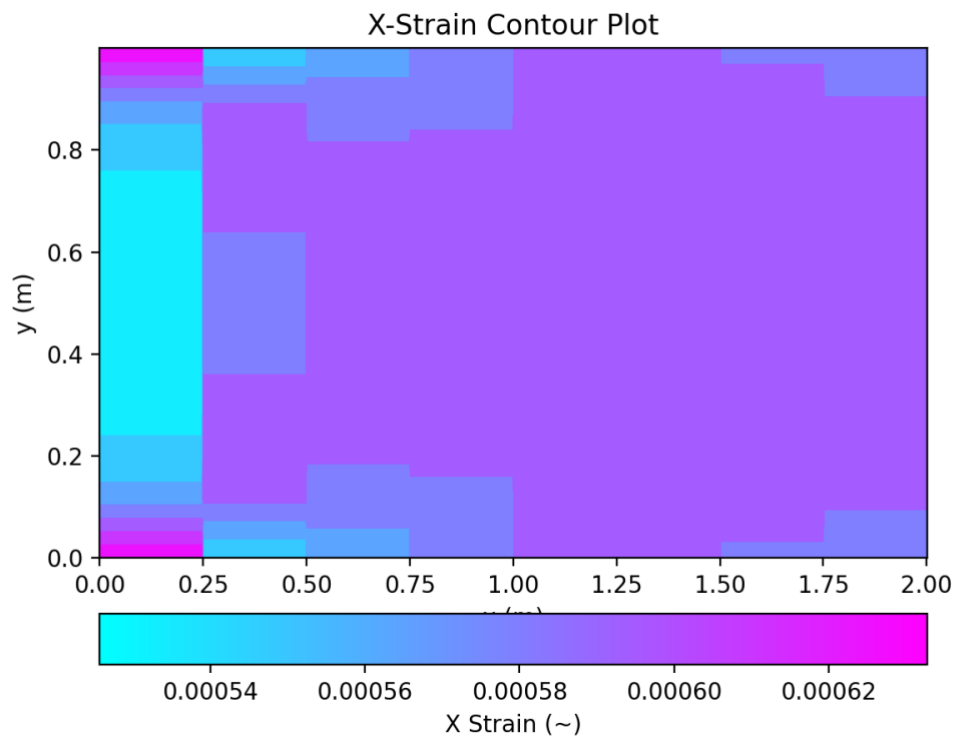
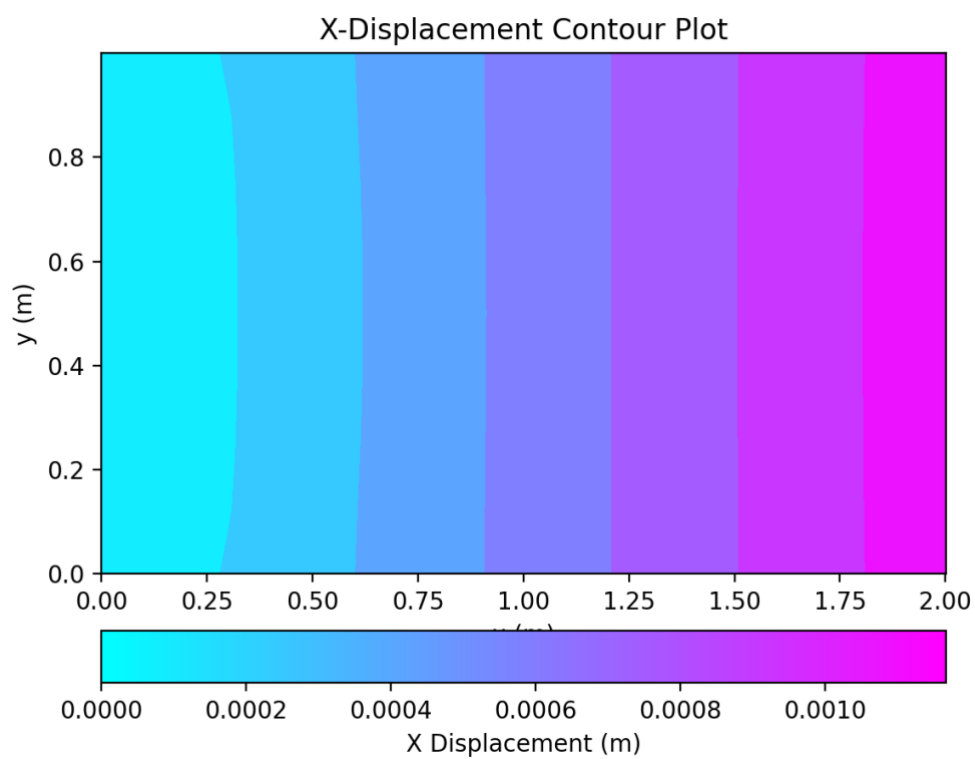


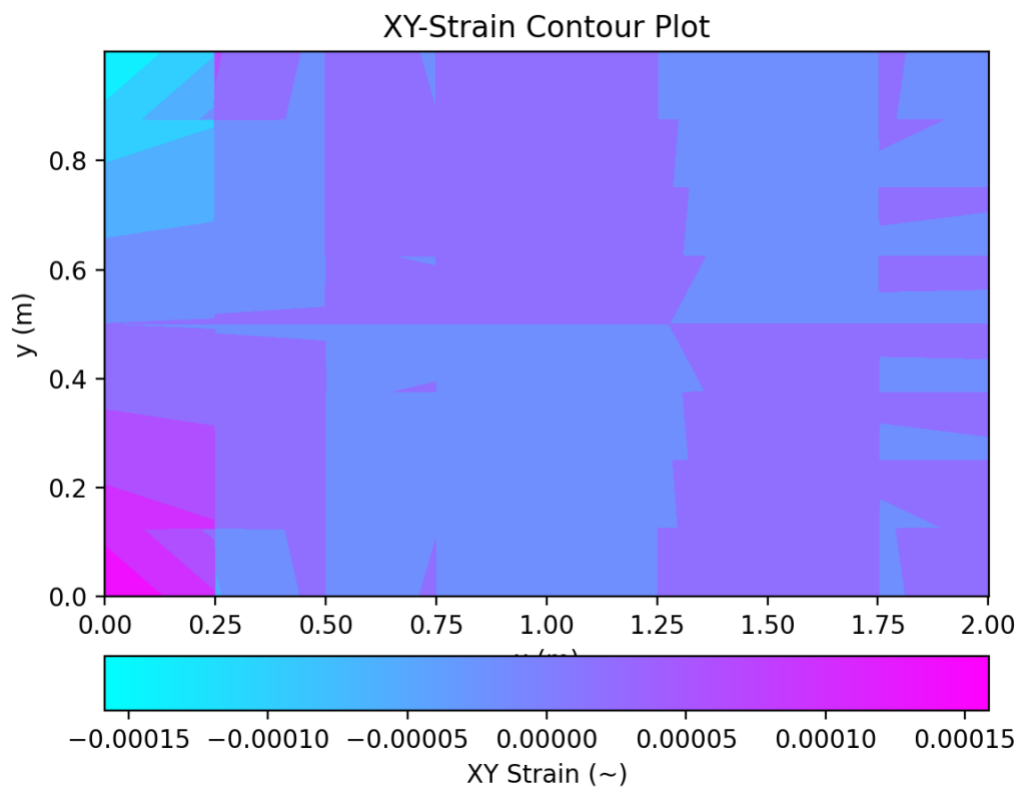
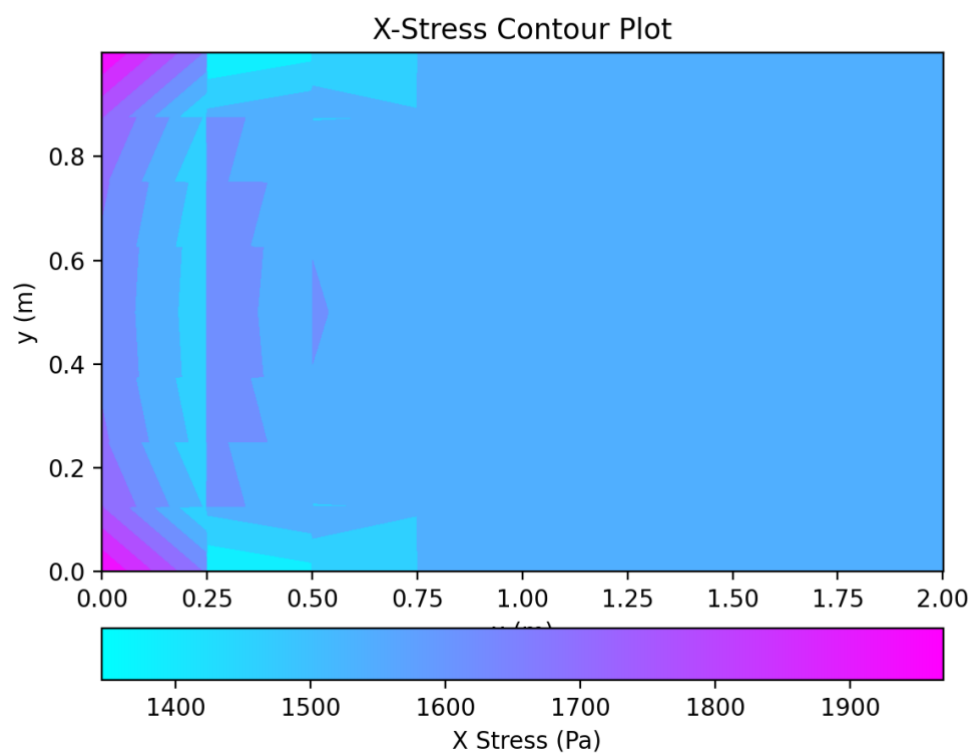
Deformation Subject to Loading (displ. magnified 200x)

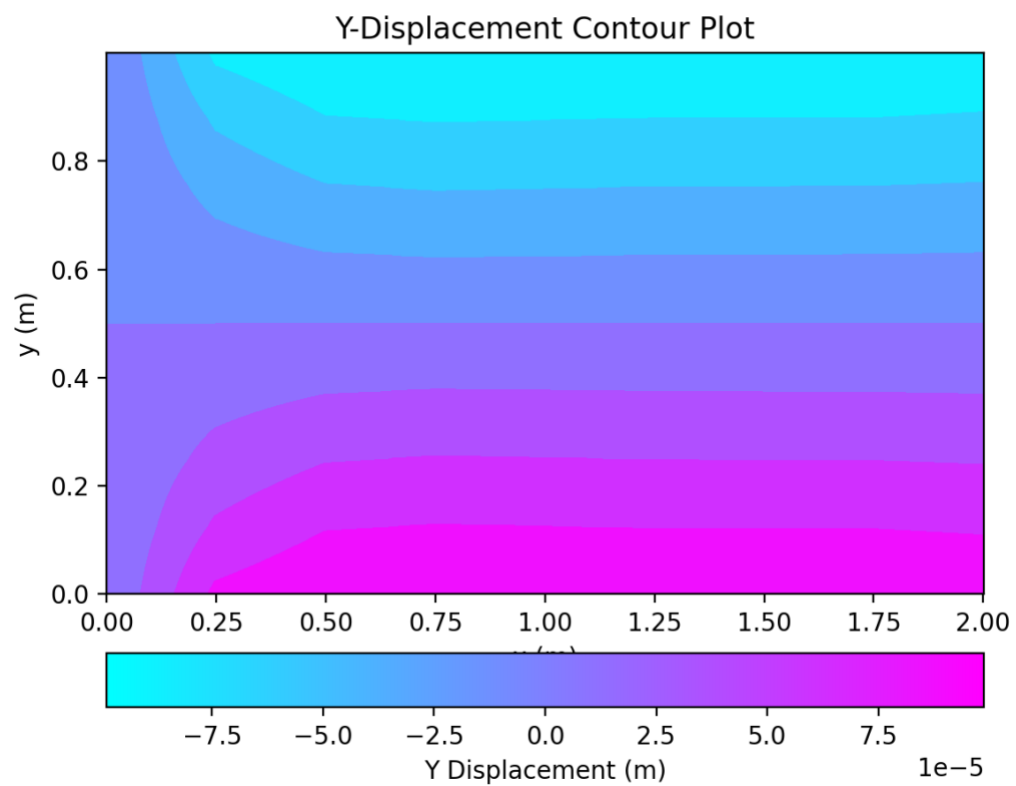
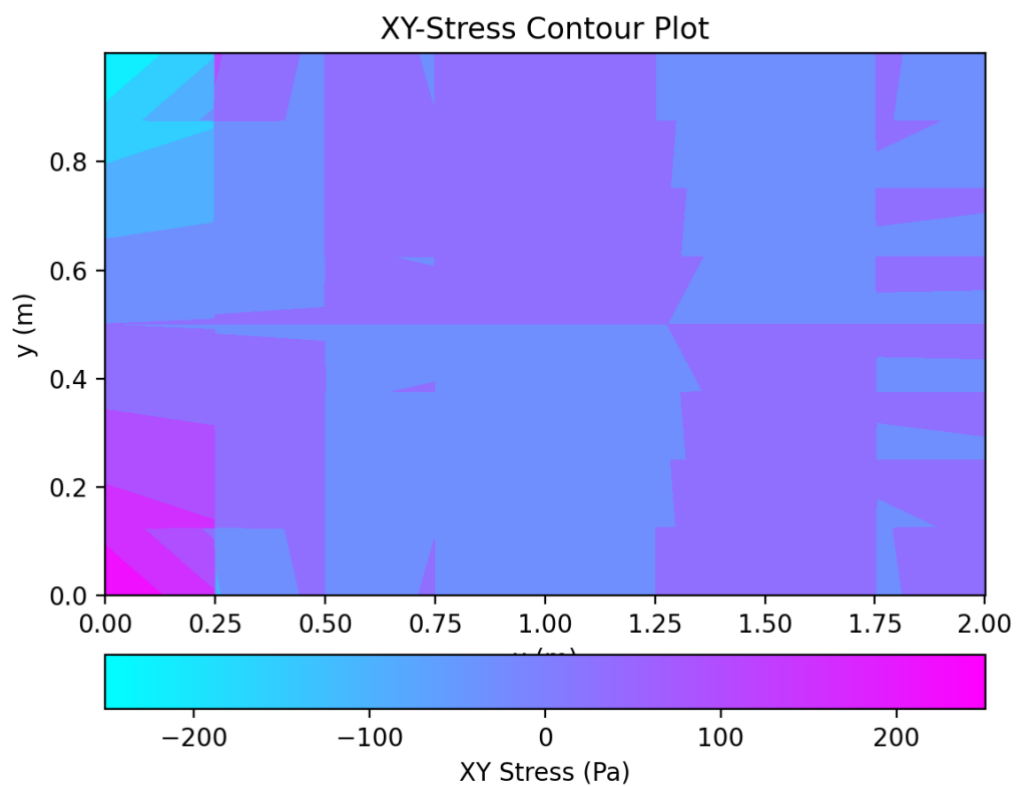


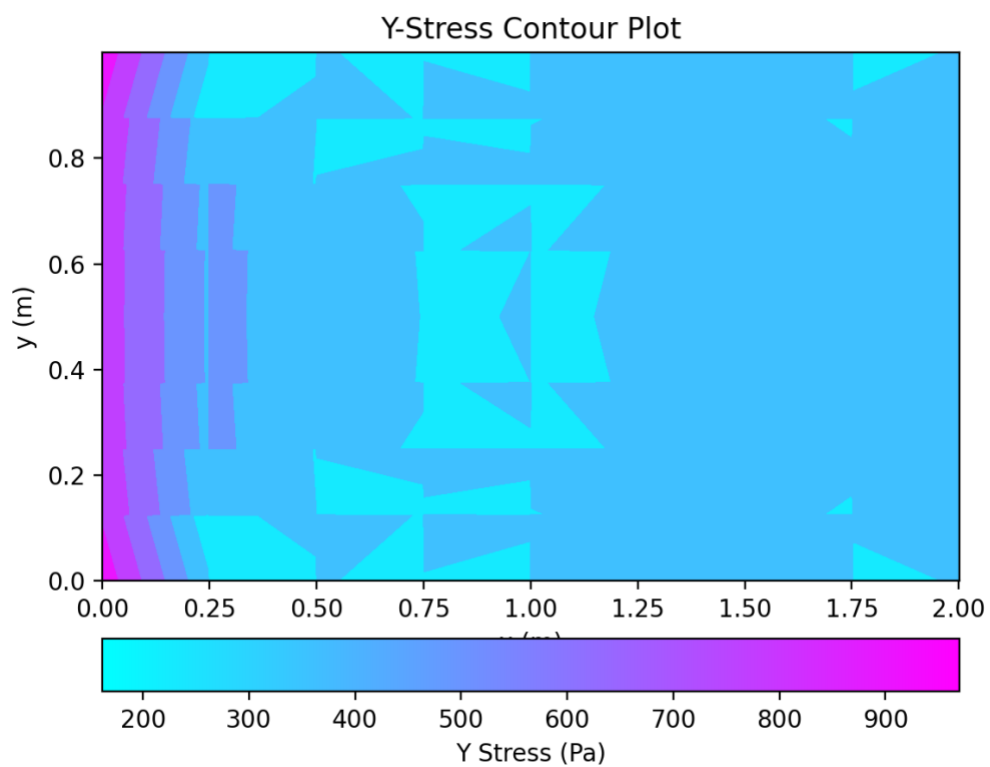
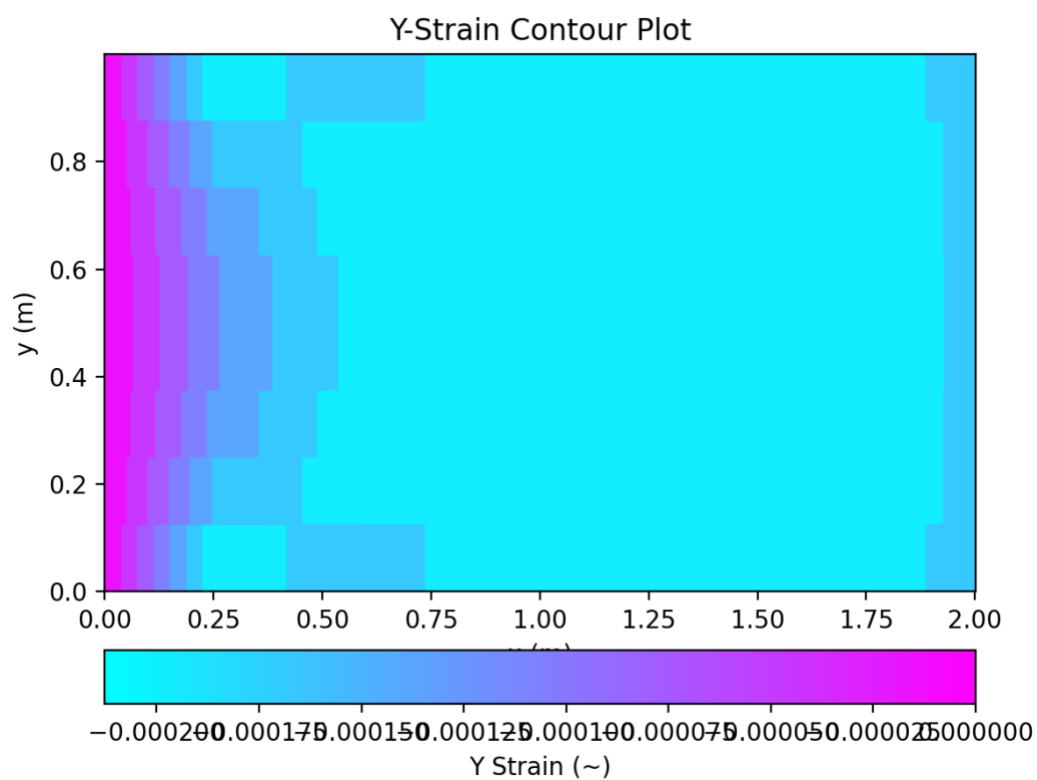
Mesh 4: 8x8

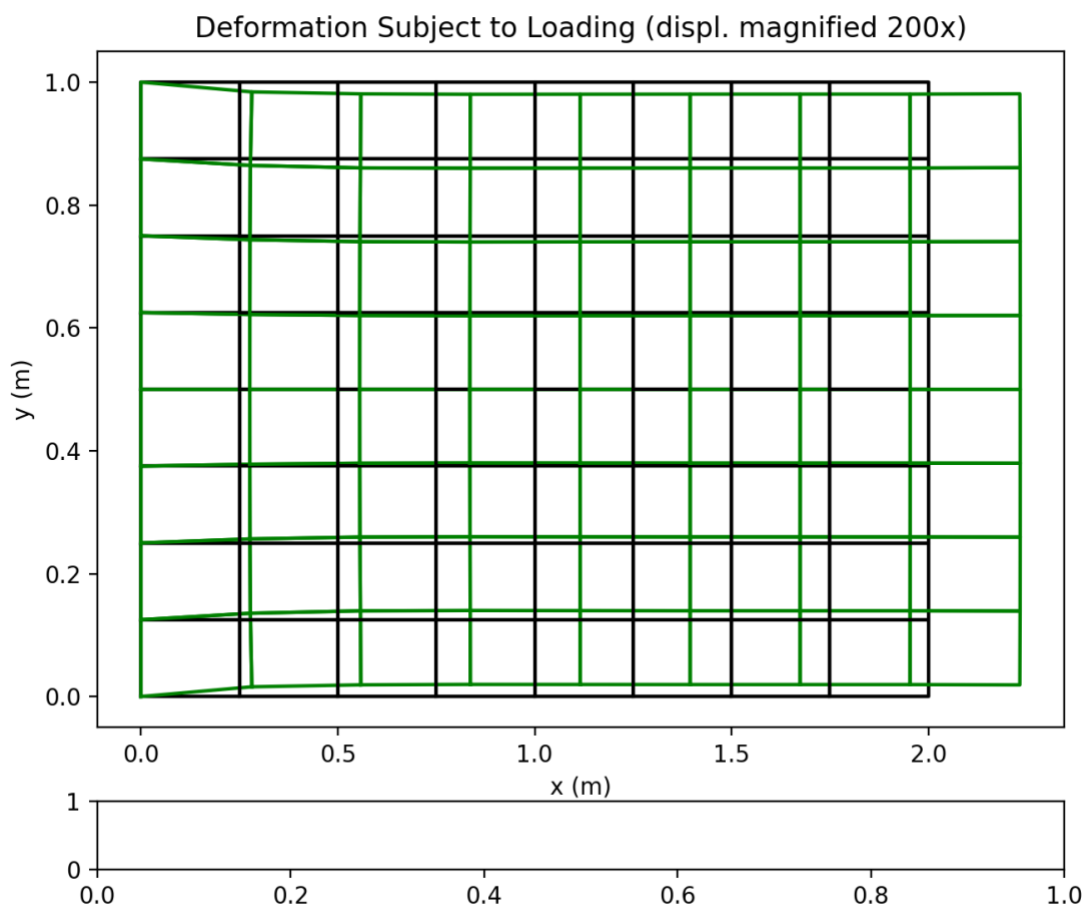




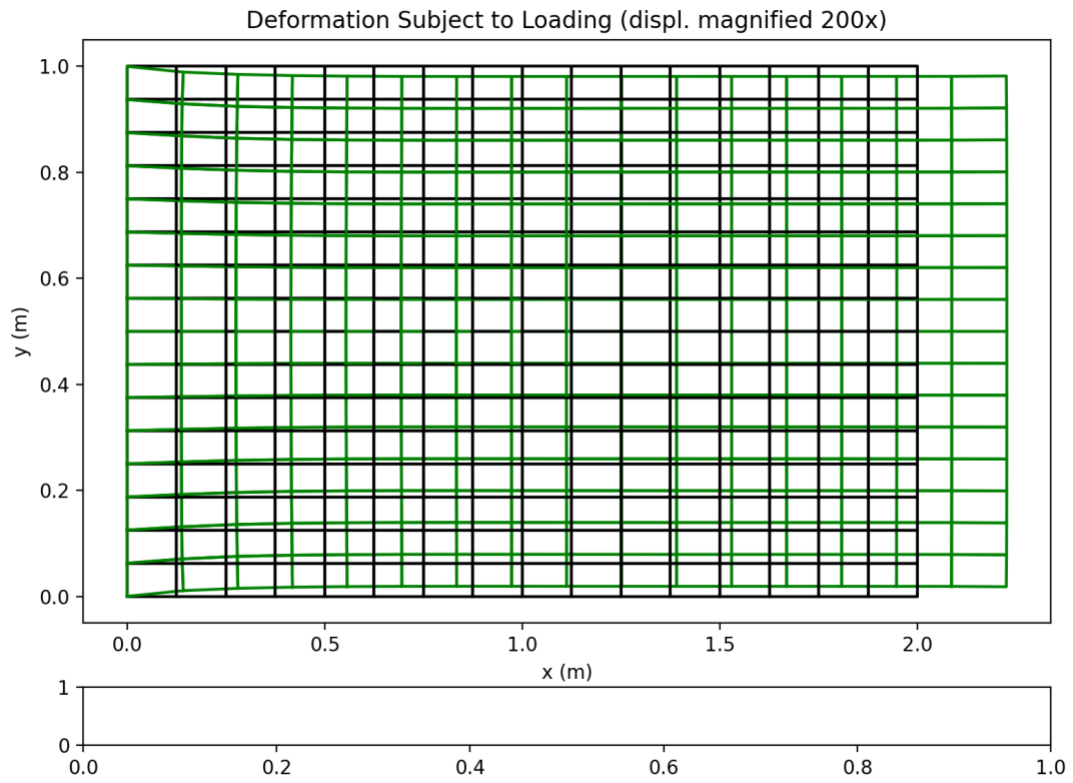


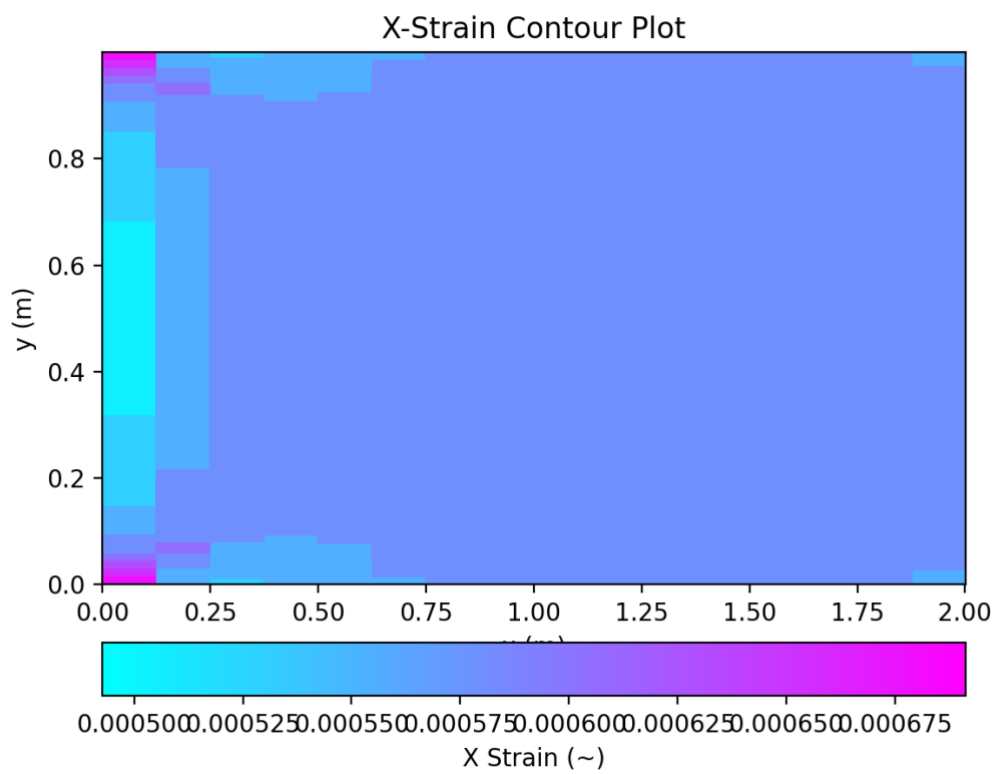
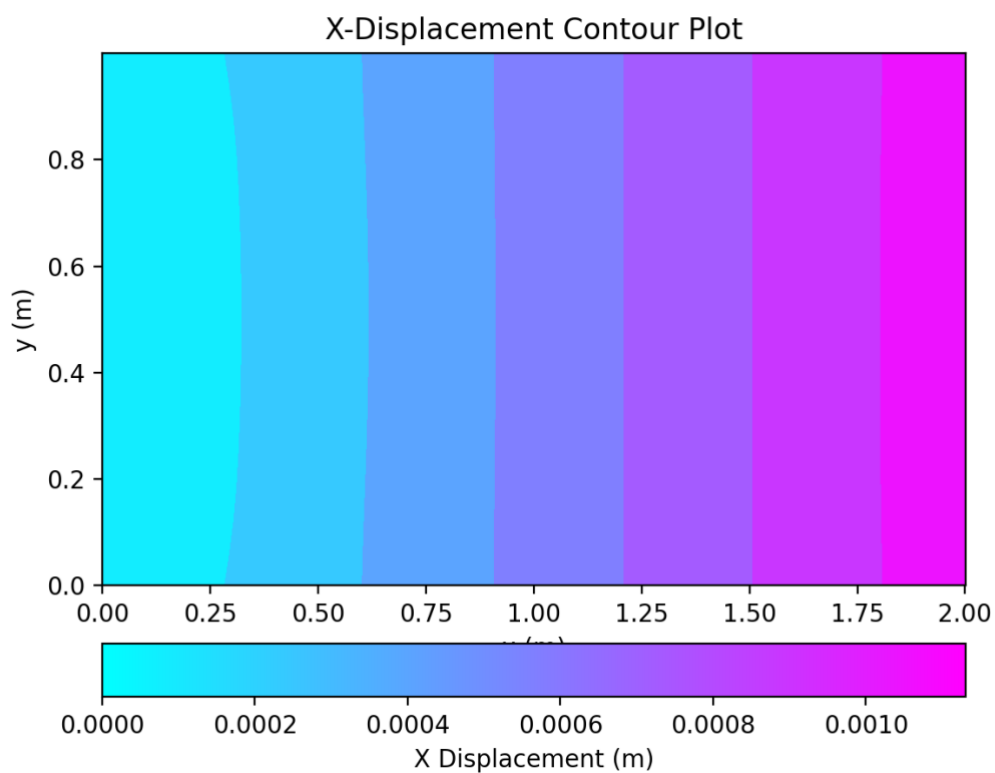


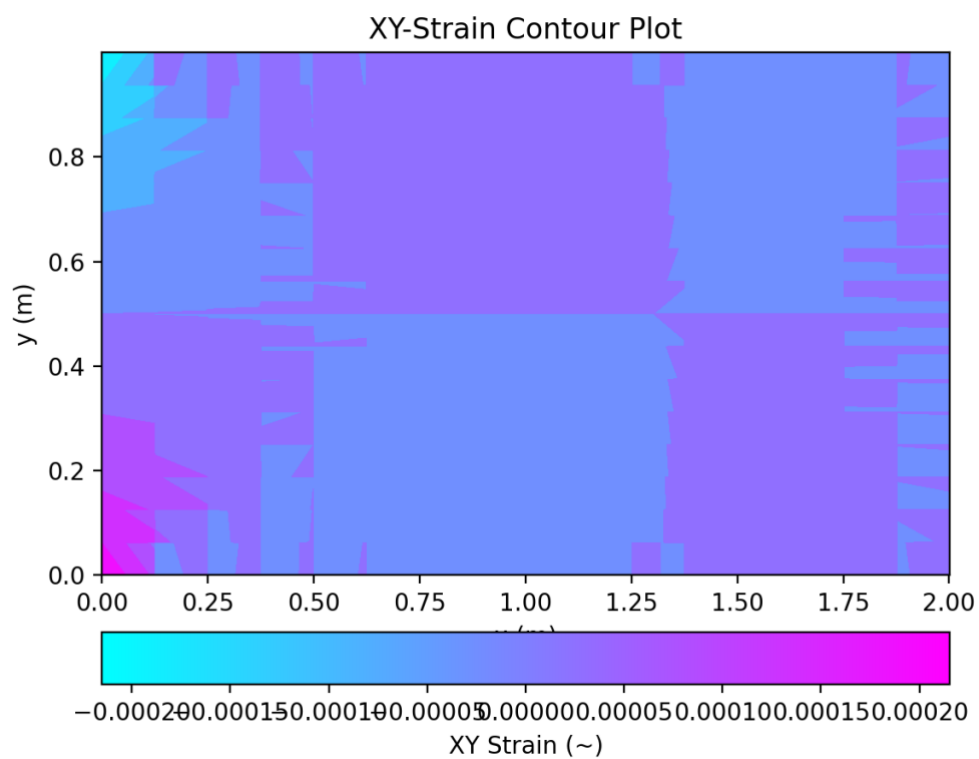
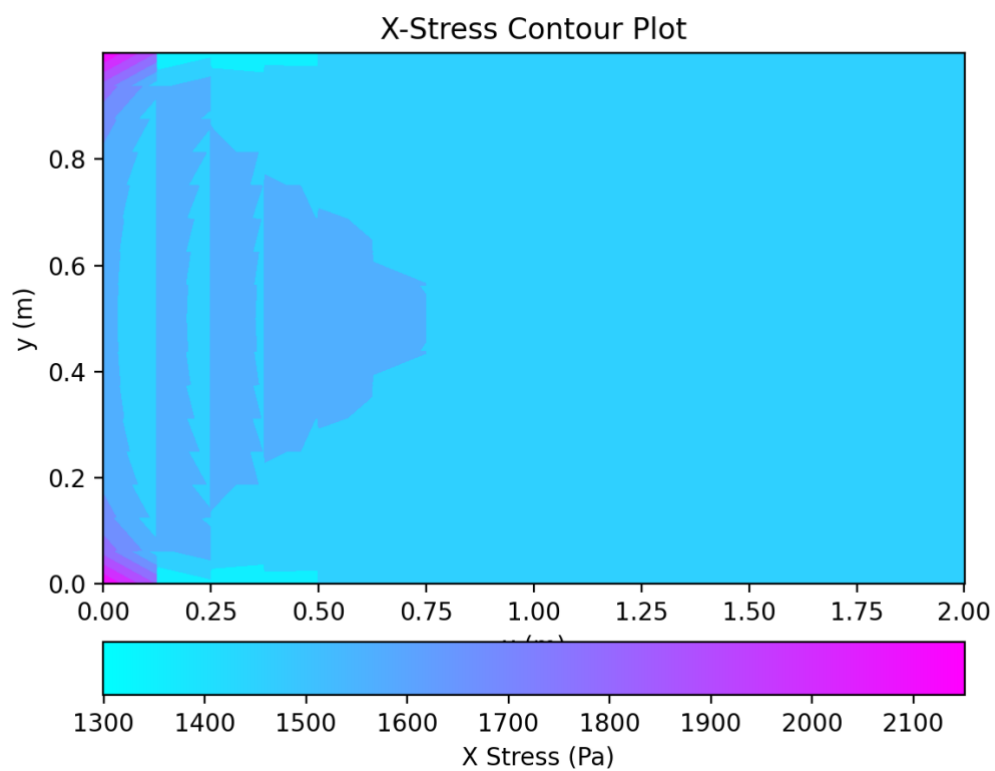


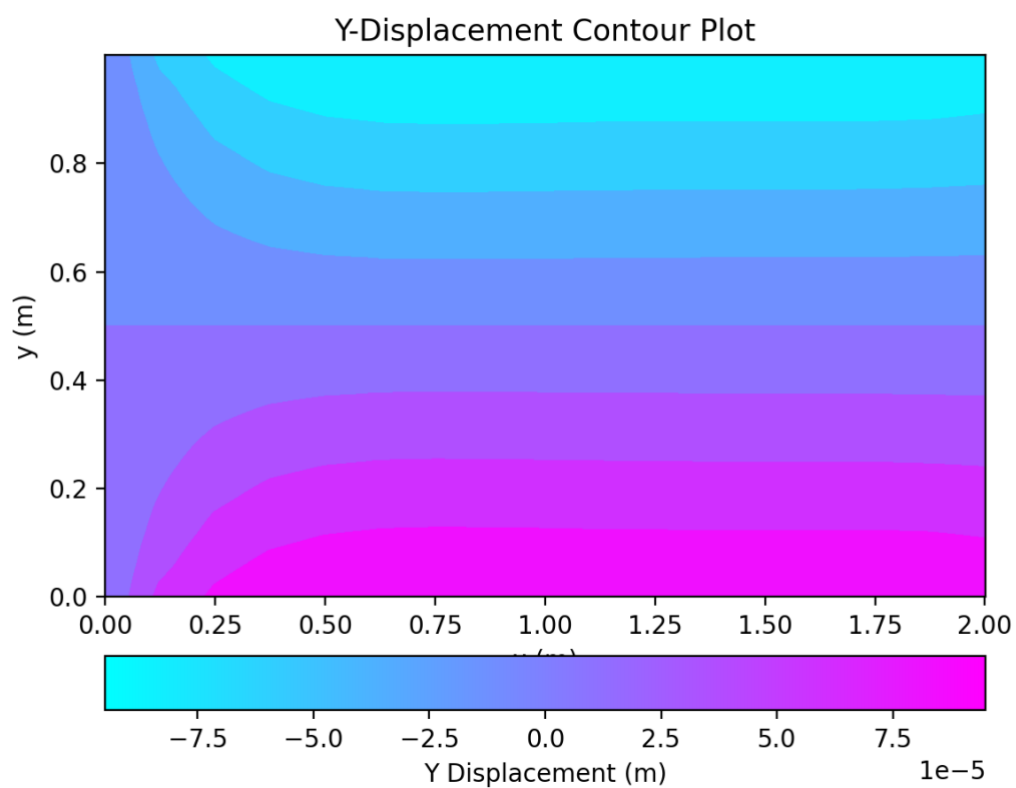
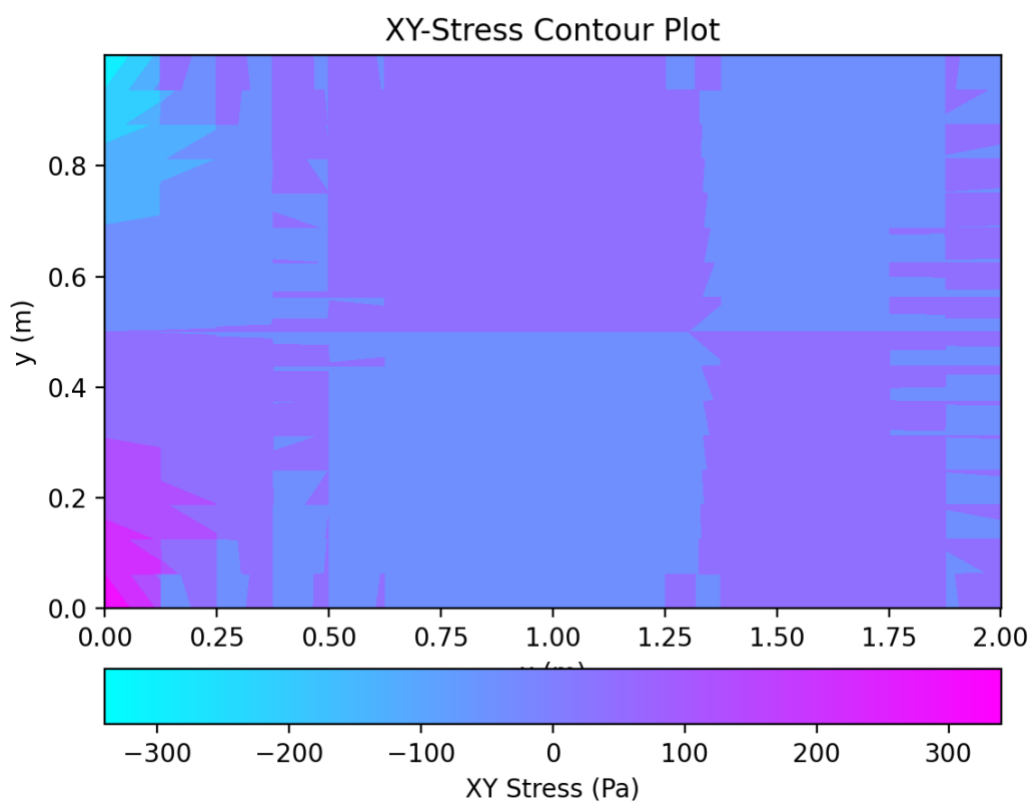


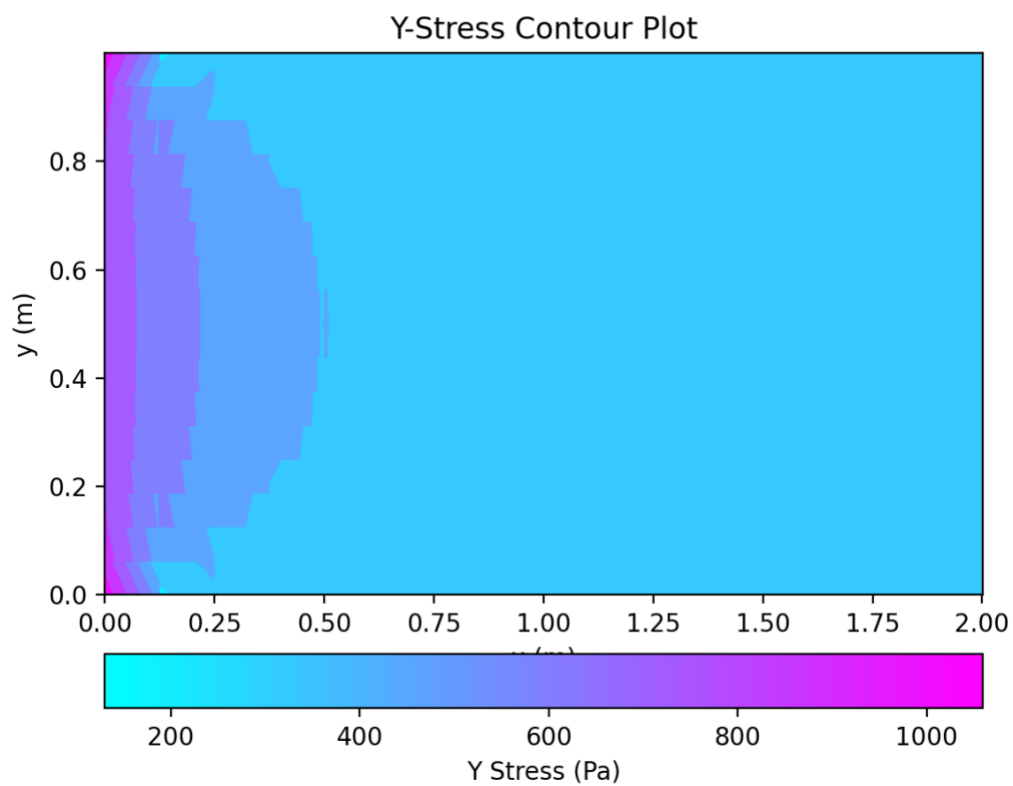
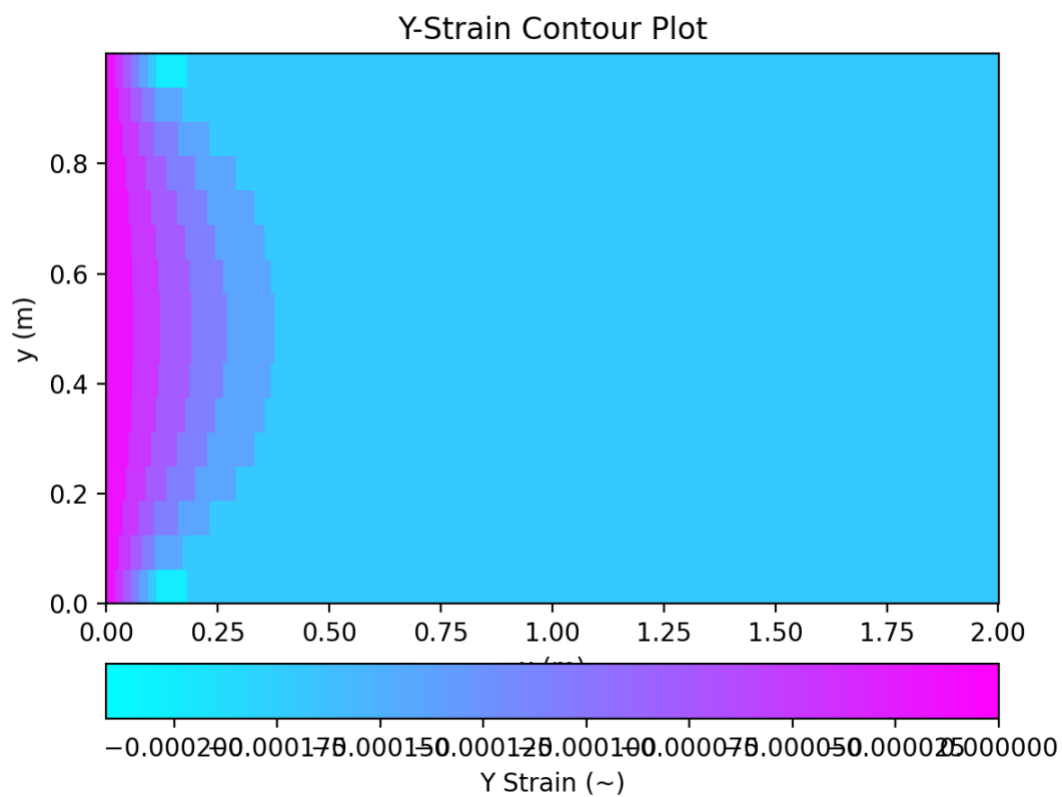
Mesh 5: 16x16

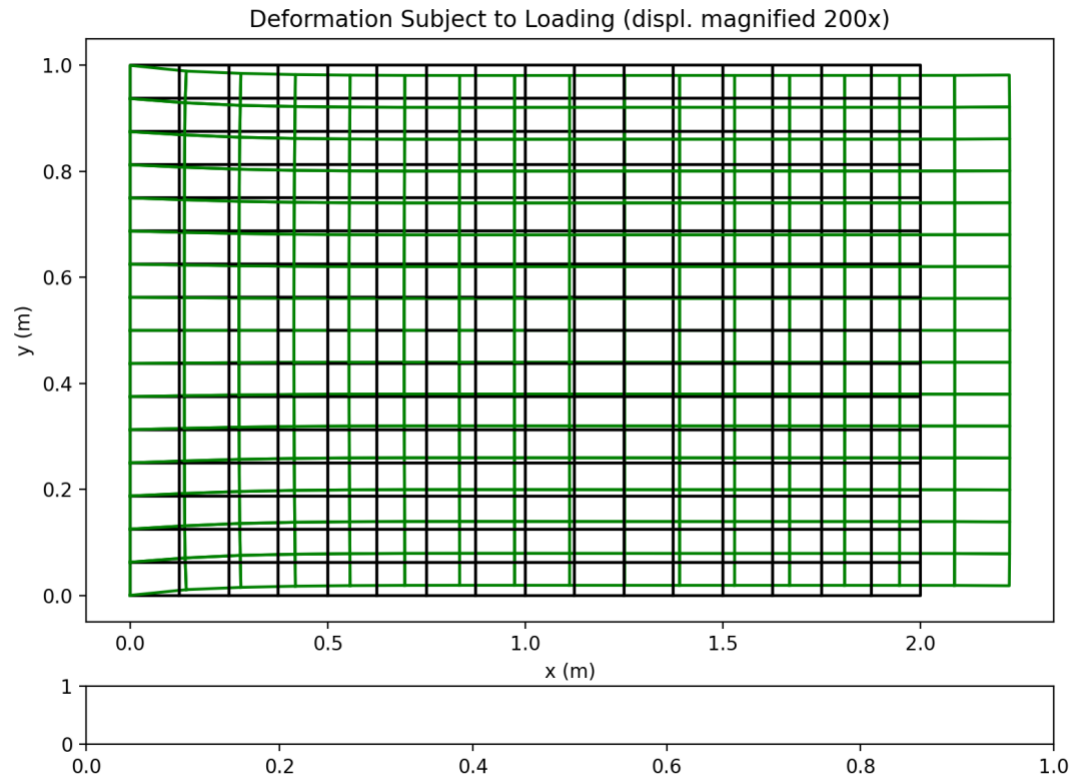




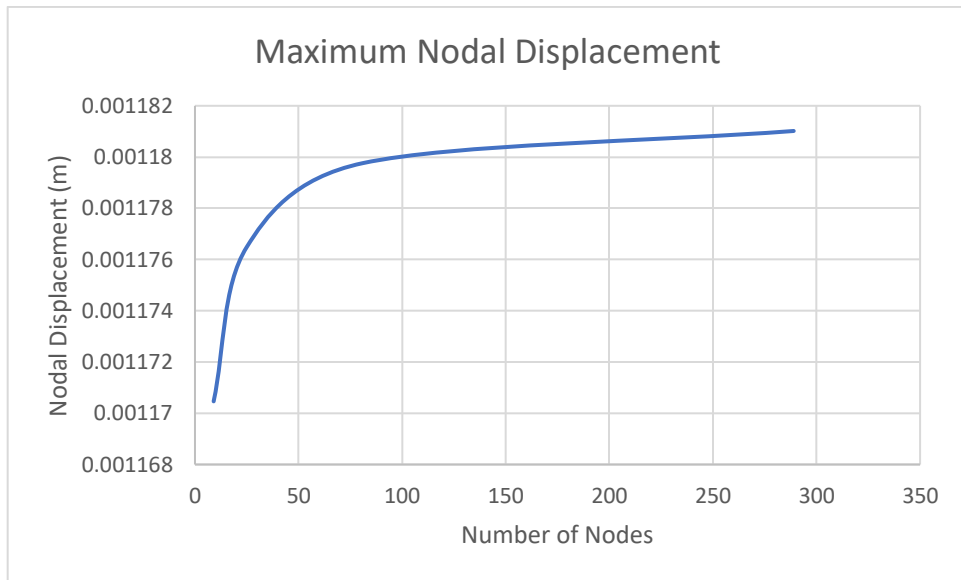
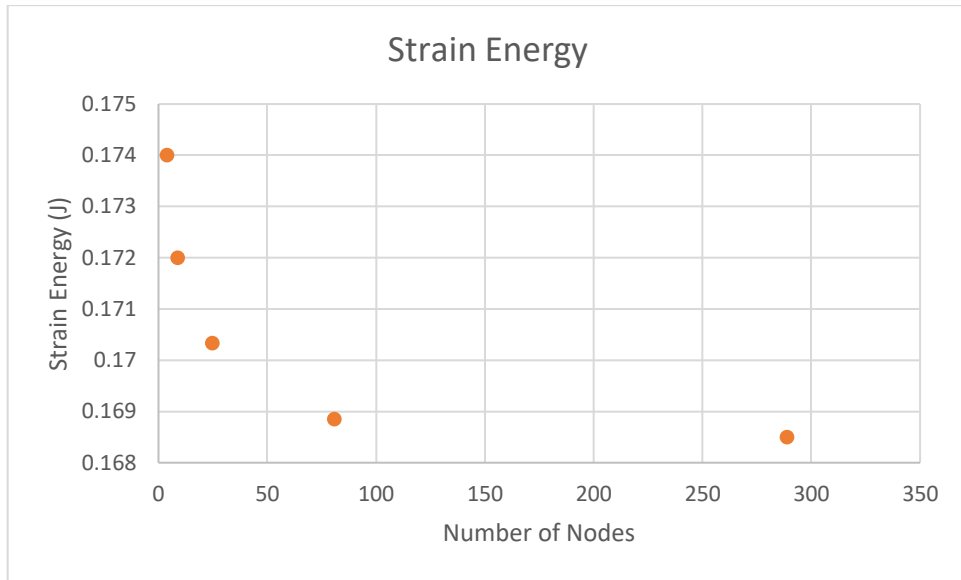


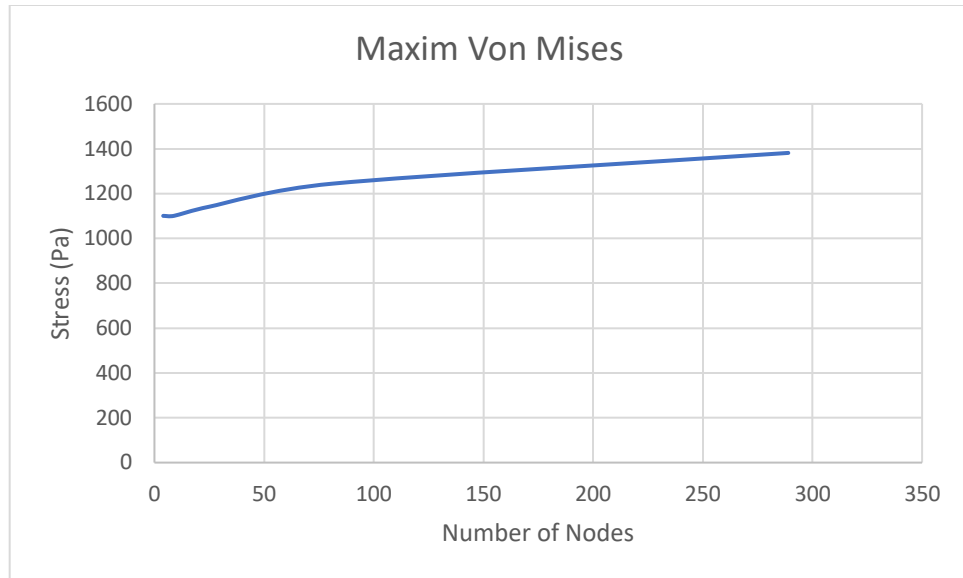






Convergence data:





The graphs show that increasing the number of nodes decreases the beam's total strain energy, suggesting convergence toward the exact solution. However, the rate of change diminishes as the mesh becomes finer, indicating progressively smaller improvements in accuracy. Similarly, coarse meshes underestimate the maximum nodal displacement, while further refinement beyond a certain point provides little improvement and increases computational time. The maximum von Mises stress follows a comparable trend, with coarse meshes producing lower stress estimates. **Underestimating stress can create a safety concern**, as it may overestimate the beam's load-carrying capacity. Therefore, sufficient mesh refinement is necessary to obtain reliable displacement and stress predictions.